Proofs of the Conjectures and Questions of Kozma and Nitzan on Bernoulli Percolation

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Abstract

Kozma and Nitzan conjectured inequalities comparing the probability that one vertex of a finite weighted graph is joined to another with the probabilities that a given finite set of vertices is joined to each of them, and they proved that the weakest of these implies that critical Bernoulli bond percolation on the hypercubic lattice has no infinite open cluster in every dimension at least two. Their paper contains nine numbered statements. Eight of them are proved here, five in a form stronger than the printed one. The universal, every-minimiser interpretation of the ninth is answered negatively when the minimum is tied; both that interpretation and the strict one are answered affirmatively when the given set has at most two vertices, and the strict one is open beyond that. Everything rests on one inequality. It bounds the mean of an increasing function of the open cluster of a vertex from below by the mean of the same function at the first vertex of the given set that the cluster reaches, and it is proved by removing the vertices of that set one at a time. Two features carry all the strength. The inequality may be conditioned on the cluster missing an auxiliary set of vertices, and a finite set of vertices may replace the single starting vertex, which turns it into a statement about a graph with one edge forced open.

1 Introduction

Let every unordered pair of vertices of a finite vertex set be open independently with a prescribed probability, and write $x \leftrightarrow y$ when the two vertices are joined by a path of open pairs. Kozma and Nitzan [KN24] asked how the probability that a vertex o is joined to a vertex b compares with the probabilities that o reaches a prescribed finite set A and that the vertices of A reach b. A path from o to A and a path from a vertex of A to b should combine into a path from o to b, but the two paths are not independent, and no proof of the resulting inequality was known. Their paper states five numbered conjectures and four numbered questions of this shape and proves that the weakest of the conjectures implies that critical Bernoulli bond percolation on the hypercubic lattice of dimension at least two has no infinite open cluster.

Conjectures 1, 2, 3, 4 and 6 and Questions 5, 7 and 9 are proved here, five of them in a form stronger than the printed one: Conjecture 2 in its pre-FKG form (1), Conjecture 4 with the vertex realising the minimum prescribed in advance, Conjecture 6 without either of the two hypotheses printed with it, Question 5 with the coefficient vector bounding the probability that the starting vertex reaches both its target and the given set, and Question 9 for every vertex realising its minimum. Question 8 is answered negatively when the vertex realising the minimum is not unique, by the four-vertex example of Section 9; it is answered affirmatively when the given set has at most two vertices, and it is open for three or more.

Everything rests on one inequality. Order the vertices of a finite set T by the mean of an increasing function F of the open cluster, and let J_x be the event that x is the first vertex of T that the open cluster of o reaches. The inequality asserts

$$\sum_{x\in T} \mathbb{P}(J_x)\,\mathbb{E}\big[F(C_x)\big] \le \mathbb{E}\big[F(C_o);\, o \leftrightarrow T\big] \;:$$

the mean of F at o, restricted to the event that o reaches T, is at least the mean of F at the vertex through which T is first reached. For the indicator of the event that b lies in the cluster this is a statement about connection probabilities, and it is the inequality from which Conjecture 3 was derived.

Two extensions of it carry the present note. The first conditions every expectation on the open cluster missing a prescribed set Y of vertices, Theorem 4.1. Applied with Y the single vertex a at which the mean of F is smallest, and applied not to F but to the function Φ_a of Section 5, which subtracts from $F(K)$ the mean of F at a in a configuration in which all pairs meeting K are closed, it yields Conjecture 4 with the vertex a prescribed in advance, Theorem 5.2, and with it Conjectures 1, 2 and 3 and Questions 5 and 7.

The second extension replaces the single vertex o by a finite set R of vertices and the cluster C_o by the union of the clusters of the vertices of R, Theorem 7.16. For a set of two vertices that union, and the events that it misses a set and that it meets a set, do not depend on the state of the pair joining the two vertices. The inequality therefore holds verbatim in the graph in which that pair is forced open, and Conjecture 6 follows. Question 9 follows from the same theorem with the random set obtained by exposing the pairs at o.

Sections 2 and 3 fix the model, state the nine numbered items and record the inputs; Section 4 proves Theorem 4.1; Sections 5 to 9 deduce the nine items in the order 4, 2, 1, 3, 7, 5, 6, 9, 8; and Section 10 records which results are kernel-checked and which extensions to a set of vertices are proved only here.

2 The Model and the Nine Numbered Statements

Let V be a finite set and let every unordered pair e of elements of V be open with probability $p_e \in [0,1]$, independently over the pairs. A pair of weight zero is an absent edge, so this is percolation on a finite weighted graph with vertex set V. Write $\mathbb{P}$ and $\mathbb{E}$ for the resulting probability and expectation. For vertices x and y, write $x \leftrightarrow y$ when there is a path of open pairs from x to y, and $x \not\leftrightarrow y$ otherwise; connection is reflexive, so $x \leftrightarrow x$ always. Write

$$C_x = \{y \in V : \ x \leftrightarrow y\}$$

for the open cluster of x and $\mathcal{C}_x$ for the set of open pairs having at least one endpoint in C_x. For a set K of pairs let $\operatorname{span} K$ be the set of vertices lying in a pair of K, so that $C_x = \{x\} \cup \operatorname{span} \mathcal{C}_x$. For a set B of vertices put $C_B = \bigcup_{y \in B} C_y$, write $\mathcal{C}_B$ for the set of open pairs having at least one endpoint in C_B, and write $\{x \leftrightarrow B\}$ for the event $\{C_x \cap B \neq \varnothing\}$ and $\{x \not\leftrightarrow B\}$ for its complement. The indicator of an event is written $\mathbf{1}_A$ when the event has been given the name A, and $\mathbf{1}\{\,\cdot\,\}$ when it is written out. A function F from subsets of V to the real numbers is increasing if $F(K) \leq F(L)$ whenever $K \subseteq L$. Every set in this note is finite, and A is always a nonempty subset of V.

Kozma and Nitzan write $\mathbb{P}(0 \leftrightarrow b)$ where we write $\mathbb{P}(o \leftrightarrow b)$. They call a function f of a vertex and a configuration a monotone cluster property if $f(v, \omega)$ depends only on the open cluster of v and is increasing in it, and if $f(v, \omega) = f(w, \omega)$ whenever the pair $\{v, w\}$ is open. Their five conjectures and four questions are the following. There is no Conjecture 5 and no Question 6.

Conjecture 2.1 (Kozma and Nitzan, display (1))**.** Let o and b be vertices and let A be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b) \geq \mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b)\,.$$

The second conjecture replaces the product of two probabilities by the probability of an intersection, which the Harris inequality makes formally weaker.

Conjecture 2.2 (Kozma and Nitzan, display (2))**.** Let o and b be vertices and let A be a nonempty set of vertices. Then

$$\mathbb{P}(o \leftrightarrow b) \geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b)\,.$$

Kozma and Nitzan call Conjecture 2.1 the post-FKG conjecture and Conjecture 2.2 the pre-FKG conjecture, and they record the formally stronger form

$$\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \geq \min_{x \in A} \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b)\,, \tag{1}$$

their display (3).

Conjecture 2.3 (Kozma and Nitzan, page 15)**.** For every $\varepsilon > 0$ there is a $\delta > 0$ such that, for every finite weighted graph, every nonempty set A of vertices and all vertices o and b with $\mathbb{P}(o \leftrightarrow A) > 1 - \delta$ and with $\mathbb{P}(x \leftrightarrow b) > 1 - \delta$ for every $x \in A$, one has $\mathbb{P}(o \leftrightarrow b) > 1 - \varepsilon$.

Conjecture 2.4 (Kozma and Nitzan, page 32)**.** Let f be a monotone cluster property, let A be a nonempty set of vertices and let o be a vertex. Then

$$\mathbb{E}\big[f(o, \cdot);\, o \leftrightarrow A\big] \geq \min_{x \in A} \mathbb{E}\big[f(x, \cdot);\, o \leftrightarrow A\big]\,.$$

For a pair e, write $\mathbb{P}^e$ and $\mathbb{E}^e$ for the probability and the expectation in the percolation measure in which the weight of e is replaced by one and all other weights are unchanged. Kozma and Nitzan write $G \downarrow e$ for the corresponding graph, which is the contraction of e as far as connection events are concerned.

Conjecture 2.5 (Kozma and Nitzan, displays (39) and (40))**.** Let A be a nonempty set of vertices, let b be a vertex, let $a \in A$ minimise $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$, and let $e = \{v, w\}$ be a pair. Assume

$$\mathbb{P}(x \leftrightarrow b) \geq \mathbb{P}(x \leftrightarrow A)\,\mathbb{P}(a \leftrightarrow b) \qquad \text{for } x = v \text{ and } x = w\,. \tag{2}$$

Then

$$\mathbb{P}^e(v \leftrightarrow b) \geq \mathbb{P}^e(v \leftrightarrow A)\,\mathbb{P}^e(a \leftrightarrow b)\,. \tag{3}$$

Question 2.6 (Kozma and Nitzan, display (38))**.** Let A be a nonempty set of vertices and let o be a vertex. Are there numbers $c_x \geq 0$ for $x \in A$ with $\sum_{x \in A} c_x = 1$, not depending on b, such that

$$\mathbb{P}(o \leftrightarrow b) \geq \sum_{x \in A} c_x\, \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b) \qquad \text{for every vertex } b?$$

The last three questions all ask whether a specified vertex $a \in A$ satisfies the inequality of the paper's display (41),

$$\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \geq \mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b)\,. \tag{4}$$

Question 2.7 (Kozma and Nitzan, page 36)**.** Does (4) hold when a minimises $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$?

Question 2.8 (Kozma and Nitzan, page 36)**.** Does (4) hold when a minimises $\mathbb{P}(x \leftrightarrow b,\ o \not\leftrightarrow A)$ over $x \in A$?

Question 2.9 (Kozma and Nitzan, page 36)**.** Write $\mathbb{P}_\star$ and $\mathbb{E}_\star$ for the probability and the expectation in the percolation measure in which every pair incident to o is closed and all other weights are unchanged. Does (4) hold, with all probabilities evaluated in the original measure, when a minimises $\mathbb{P}_\star(x \leftrightarrow b)$ over $x \in A$?

The minimand in Question 2.8 is the probability of the joint event that x reaches b and that o does not reach A; it is not the minimand of (4) itself, which is treated by (1).

3 Conditional Positive Association and a Covariance Inequality

Two families of inequalities are taken as given. The first is the conditional positive association of van den Berg, Häggström and Kahn [BHK06].

Theorem 3.1 (van den Berg, Häggström and Kahn)**.** *Let s be a vertex, let X_1 and X_2 be sets of vertices not containing s and let g_1 and g_2 be nonnegative increasing functions of sets of pairs. Then*

$$\mathbb{E}\big[g_1(\mathcal{C}_s);\, s \not\leftrightarrow X_1\big]\,\mathbb{E}\big[g_2(\mathcal{C}_s);\, s \not\leftrightarrow X_2\big] \le \mathbb{E}\big[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s);\, s \not\leftrightarrow X_1 \cap X_2\big]\,\mathbb{P}(s \not\leftrightarrow X_1 \cup X_2)\,.$$

A companion inequality of the same authors governs two clusters at once, and it is what Sections 7.4 and 9 use.

Theorem 3.2 (van den Berg, Häggström and Kahn)**.** *Let S_1 and S_2 be sets of vertices, and let g_3 and g_4 be real functions of two sets of pairs, both increasing in the first argument and decreasing in the second. Then, with both functions evaluated at $(\mathcal{C}_{S_1}, \mathcal{C}_{S_2})$,*

$$\mathbb{E}\big[g_3;\, S_1 \not\leftrightarrow S_2\big]\,\mathbb{E}\big[g_4;\, S_1 \not\leftrightarrow S_2\big] \le \mathbb{P}(S_1 \not\leftrightarrow S_2)\,\mathbb{E}\big[g_3 g_4;\, S_1 \not\leftrightarrow S_2\big]\,.$$

Exchanging S_1 and S_2 shows that Theorem 3.2 also holds when g_3 and g_4 are both decreasing in the first argument and increasing in the second, and replacing g_4 by $-g_4$ shows that it reverses, to

$$\mathbb{P}(S_1 \not\leftrightarrow S_2)\,\mathbb{E}\big[g_3 g_4;\, S_1 \not\leftrightarrow S_2\big] \le \mathbb{E}\big[g_3;\, S_1 \not\leftrightarrow S_2\big]\,\mathbb{E}\big[g_4;\, S_1 \not\leftrightarrow S_2\big]\,, \tag{5}$$

when one of the two functions is increasing in the first argument and decreasing in the second while the other is decreasing in the first and increasing in the second.

Taking $X_1 = X_2$ in Theorem 3.1 gives the conditional positive association of two increasing functions of a single cluster,

$$\mathbb{E}\big[g_1(\mathcal{C}_s);\, s \not\leftrightarrow X\big]\,\mathbb{E}\big[g_2(\mathcal{C}_s);\, s \not\leftrightarrow X\big] \le \mathbb{P}(s \not\leftrightarrow X)\,\mathbb{E}\big[g_1(\mathcal{C}_s)g_2(\mathcal{C}_s);\, s \not\leftrightarrow X\big]\,. \tag{6}$$

Both sides of (6) increase by the same amount when a constant is added to g_1 or to g_2, and each takes finitely many values, so the nonnegativity assumption may be dropped in (6): monotonicity of g_1 and g_2 suffices. Taking $g_1 = g_2 = 1$ in Theorem 3.1 gives the Harris inequality for the two decreasing events $\{s \not\leftrightarrow X_1\}$ and $\{s \not\leftrightarrow X_2\}$. The Harris inequality [H60] for two increasing events is used repeatedly and without further comment. The Ahlswede and Daykin four functions theorem [AD78] is used once, in Lemma 7.11.

The second input is a family of covariance inequalities for one open cluster, conditioned on that cluster missing a prescribed set of vertices, and discounted by a finite list of auxiliary vertices. Fix a vertex s, a set Y of vertices with $s \notin Y$, a list $z_1, \ldots, z_\ell$ of distinct vertices and two further distinct vertices o and v, with all the named vertices outside Y and pairwise distinct. Write ν for the conditional law $\mathbb{P}(\,\cdot \mid s \not\leftrightarrow Y)$ and put

$$\pi_j(u) = \mathbb{P}\big(u \in C_{z_j} \,\big|\, z_j \not\leftrightarrow B_j\big), \qquad \tau = \mathbb{P}\big(o \in C_v \,\big|\, v \not\leftrightarrow B_{\ell+1}\big)\,, \tag{7}$$

where $B_1 = \{s\} \cup Y$ and $B_{j+1} = B_j \cup \{z_j\}$ for $1 \le j \le \ell$. For a real function q on V put $q^0 = q$ and

$$q^j(u) = q^{j-1}(u) - \pi_j(u)\, q^{j-1}(z_j) \quad \text{for } 1 \le j \le \ell, \qquad \mathfrak{M}[q] = q^\ell(o) - \tau\, q^\ell(v)\,. \tag{8}$$

The jth step removes from the value at every vertex the part explained by z_j, at the rate at which z_j reaches that vertex while missing everything named before it. All the constants are computed under $\mathbb{P}$ and not under ν.

Theorem 3.3. *Let s, Y, the list $z_1, \ldots, z_\ell$ and the vertices o and v be as above, let every pair weight lie strictly between zero and one, and let g be an increasing function of sets of pairs. Then*

$$\mathfrak{M}\big[\, u \mapsto \mathrm{Cov}_\nu\big(g(\mathcal{C}_s), \mathbf{1}\{u \in C_s\}\big)\,\big] \ge 0\,.$$

For $\ell = 0$ the conclusion reads

$$\mathrm{Cov}_\nu\big(g(\mathcal{C}_s), \mathbf{1}\{o \in C_s\}\big) \ge \mathbb{P}\big(o \in C_v \bigm| v \nleftrightarrow \{s\} \cup Y\big)\ \mathrm{Cov}_\nu\big(g(\mathcal{C}_s), \mathbf{1}\{v \in C_s\}\big)\,,$$

and if in addition $Y = \varnothing$ and $g = \mathbf{1}\{v \in \cdot\}$ it is the Harris inequality.

One further construction is recorded here, since it is a property of the objects appearing in the proof of Theorem 3.3 rather than of its statement, and Section 7 uses it. Condition on the set $\mathcal{C}_Y$ of open pairs met by paths from Y. On the event $D = \{s \nleftrightarrow Y\}$, the vertex set $C_Y = Y \cup \operatorname{span} \mathcal{C}_Y$ misses C_s, every pair with exactly one endpoint in C_Y is closed, and the pairs with both endpoints outside C_Y retain their original weights and are independent of the pairs meeting C_Y. Write $\bar{g}$ for the mean of $g(\mathcal{C}_s)$ in a configuration in which every pair meeting C_Y is closed and every remaining pair keeps its original weight. Then the random variable

$$\Theta = \mathbf{1}_D\big(g(\mathcal{C}_s) - \bar{g}\big) \tag{9}$$

has $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$. For a set K of vertices disjoint from $\{s\} \cup Y$, let $\Psi(K)$ be minus the mean of Θ computed in the configuration in which every pair meeting K is deleted. Then $\Psi(\varnothing) = 0$, the function Ψ is nonnegative and increasing, and for every vertex z and every set B containing $\{s\} \cup Y$ but not z,

$$\mathbb{E}\big[\Theta;\, z \nleftrightarrow B\big] = -\,\mathbb{E}\big[\Psi(C_z);\, z \nleftrightarrow B\big]\,. \tag{10}$$

Both the vanishing of the conditional mean of Θ and the identity (10) express the domain Markov property of percolation: given $\mathcal{C}_B$ for a set B of vertices, the pairs with both endpoints outside C_B retain their original weights and are independent of that data.

Two properties of the conditioned covariance itself are recorded for the same reason. For a set W of vertices write $\mathbb{P}_W$ and $\mathbb{E}_W$ for percolation in which every pair with an endpoint outside W is closed and every pair with both endpoints in W keeps its weight, and write Cov_W for the covariance under $\mathbb{P}_W$. For a set X of vertices containing s and a vertex $v \notin X$, put

$$\eta_v(W) = \mathrm{Cov}_W\big(g(\mathcal{C}_s), \mathbf{1}\{v \leftrightarrow X\}\big) \quad \text{for } v \in W, \qquad \eta_v(W) = 0 \quad \text{for } v \notin W\,. \tag{11}$$

The construction underlying Theorem 3.3 yields, for every W and every $Q \subseteq W$,

$$\mathbb{E}_W\big[\mathbf{1}\{s \nleftrightarrow Q\}\, \eta_v(W \setminus C_Q)\big]\, \mathbb{P}_W(v \nleftrightarrow X) \le \mathbb{P}_W\big(v \nleftrightarrow X \cup Q\big)\, \eta_v(W)\,, \tag{12}$$

and the left side of (12) decreases as Q grows.

We turn to the inequality from which all of the numbered statements are deduced.

4 A Lower Bound Conditioned on Missing a Fixed Set

Fix a set Y of vertices and an increasing function F of vertex sets. For a vertex x with $\mathbb{P}(x \nleftrightarrow Y) > 0$ put

$$m_x = \frac{\mathbb{E}\big[F(C_x);\, x \nleftrightarrow Y\big]}{\mathbb{P}(x \nleftrightarrow Y)}\,, \tag{13}$$

the mean of F at x given that x misses Y. Let T be a finite set of vertices disjoint from Y with $\mathbb{P}(x \nleftrightarrow Y) > 0$ for every $x \in T$, and let r be an injective real function on T such that $m_x \le m_y$ whenever $r(x) < r(y)$. Such a function exists: order T by the values m_x and break ties arbitrarily. For $x \in T$ and a vertex o write

$$J_x = \{o \nleftrightarrow Y\} \cap \{o \leftrightarrow x\} \cap \bigcap_{y \in T,\ r(y) < r(x)} \{o \nleftrightarrow y\} \tag{14}$$

for the event that o misses Y and that x is the vertex of smallest value of r that the open cluster of o reaches in T. The events J_x for $x \in T$ are disjoint and their union is $\{o \nleftrightarrow Y\} \cap \{o \leftrightarrow T\}$. Put

$$\Lambda_o(T) = \mathbb{E}\big[F(C_o);\, o \nleftrightarrow Y,\ o \leftrightarrow T\big] - \sum_{x \in T} \mathbb{P}(J_x)\, m_x\,. \tag{15}$$

On J_x the vertex x has the smallest value of r among the vertices of T in C_o, hence, by the assumption on r, the smallest value of m, and on J_x the clusters of o and x coincide. Consequently (15) admits the expression

$$\Lambda_o(T) = \mathbb{E}\Big[\mathbf{1}\{o \nleftrightarrow Y,\, o \leftrightarrow T\}\Big(F(C_o) - \min_{x \in T \cap C_o} m_x\Big)\Big]\,, \tag{16}$$

which does not refer to r. Two positions of o are settled by (16). If $o \in Y$ then the event $\{o \nleftrightarrow Y\}$ is empty and $\Lambda_o(T) = 0$. If $o \in T$ then $o \in T \cap C_o$ on $\{o \nleftrightarrow Y\}$, so the minimum in (16) is at most m_o, whence $\Lambda_o(T) \geq \mathbb{E}[F(C_o) - m_o;\, o \nleftrightarrow Y] = 0$ by the definition (13) of m_o.

Theorem 4.1. *Let V be a finite vertex set with pair weights in $[0,1]$, let $Y \subseteq V$, let F be an increasing function of subsets of V, let $T \subseteq V \setminus Y$ satisfy $\mathbb{P}(x \nleftrightarrow Y) > 0$ for every $x \in T$, let r be injective on T with $m_x \leq m_y$ whenever $r(x) < r(y)$, and let $o \in V$. Then $\Lambda_o(T) \geq 0$, that is*

$$\sum_{x \in T} \mathbb{P}(J_x)\, m_x \leq \mathbb{E}\big[F(C_o);\, o \nleftrightarrow Y,\ o \leftrightarrow T\big]\,.$$

For $Y = \varnothing$ this is the inequality quoted in the introduction. No sign hypothesis on F is needed. The proof removes from T the vertex at which m is largest and transfers the resulting deficit from that vertex back to o. The transfer step is the following inequality, which is what Theorem 3.3 supplies.

Proposition 4.2. *Let V be a finite vertex set with pair weights in $[0,1]$, let $Y \subseteq V$, let F be an increasing function of subsets of V, let $T \subseteq V \setminus Y$ satisfy $\mathbb{P}(x \nleftrightarrow Y) > 0$ for every $x \in T$, let r be injective on T with $m_x \leq m_y$ whenever $r(x) < r(y)$, and let $v \in V$ lie outside $Y \cup T$. Then, for every vertex o,*

$$\mathbb{P}\big(v \nleftrightarrow Y \cup T,\ o \leftrightarrow v\big)\, \Lambda_v(T) \leq \mathbb{P}\big(v \nleftrightarrow Y \cup T\big)\, \Lambda_o(T)\,.$$

Proof. We divide the proof into three steps: first an induction on the cardinality of T, then the specialisation of that induction to an empty list of auxiliary vertices, then the passage from weights strictly inside the unit interval to weights in $[0,1]$.

Assume first that every pair weight lies strictly between zero and one and that o lies outside $Y \cup T$ and differs from v; the remaining positions of o are treated in the second step. Let $z_1, \dots, z_\ell$ be distinct vertices outside $Y \cup T \cup \{o, v\}$ and let $\mathfrak{M}$ be the functional (8) built from the constants (7) with Y replaced by $Y \cup T$ at every step, so that z_j is conditioned to miss $Y \cup T \cup \{z_1, \dots, z_{j-1}\}$ and the final constant is $\tau = \mathbb{P}(o \in C_v \mid v \nleftrightarrow Y \cup T \cup \{z_1, \dots, z_\ell\})$. We start by showing, by induction on the cardinality of T and simultaneously for every such list and every pair of vertices o and v, that

$$\mathfrak{M}\big[\, u \mapsto \Lambda_u(T)\,\big] \geq 0\,. \tag{17}$$

In the base case $T = \varnothing$ each $\Lambda_u(\varnothing)$ vanishes. For the inductive step, let T be nonempty, let $k \in T$ have the largest value of r, and put $T_0 = T \setminus \{k\}$. Abbreviate

$$P = \mathbb{P}(k \nleftrightarrow Y \cup T_0), \qquad P(u) = \mathbb{P}(k \nleftrightarrow Y \cup T_0,\ u \leftrightarrow k),$$

$$I = \mathbb{E}\big[F(C_k);\, k \nleftrightarrow Y \cup T_0\big], \qquad I(u) = \mathbb{E}\big[F(C_k);\, k \nleftrightarrow Y \cup T_0,\ u \leftrightarrow k\big]\,.$$

We first record an identity. Since k has the largest value of r, adding k to T_0 leaves the events J_x for $x \in T_0$ unchanged and creates the single new event $\{u \nleftrightarrow Y\} \cap \{u \leftrightarrow k\} \cap \{u \nleftrightarrow T_0\}$, which is $\{k \nleftrightarrow Y \cup T_0,\ u \leftrightarrow k\}$: indeed on $\{u \leftrightarrow k\}$ the clusters of u and k coincide. On that event $F(C_u) = F(C_k)$, so

$$\Lambda_u(T) = \Lambda_u(T_0) + I(u) - P(u)\, m_k\,. \tag{18}$$

The last two terms on the right are handled together. Write $\mathrm{Cov}^*(u)$ for $P\,I(u) - I\,P(u)$, the conditioned covariance of Theorem 3.3 written without denominators, for the vertex k, the set $Y \cup T_0$ and the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$, and put $\gamma_k = m_k P - I$. Expanding both sides gives

$$P\big(I(u) - P(u)\, m_k\big) = \mathrm{Cov}^*(u) - \gamma_k\, P(u)\,. \tag{19}$$

Theorem 3.3 for the vertex k, the set $Y \cup T_0$, the same list and the same two vertices gives $\mathfrak{M}[\mathrm{Cov}^*] \geq 0$. Theorem 3.3 for the same vertex and set and for the increasing function $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \varnothing\}$ gives $\mathfrak{M}[P] \geq 0$. Indeed $\mathfrak{M}$ evaluates its argument only at o, at v and at the vertices of the list, all of which differ from k; on $\{u \leftrightarrow k\}$ with $u \neq k$ the cluster of k contains an open pair, and on $\{k \not\leftrightarrow Y \cup T_0\}$ the mean of that indicator is P minus the probability that k misses $Y \cup T_0$ and lies in no open pair, so the covariance in question is the latter probability times $P(u)$, and that probability is positive because all weights are strictly below one. Finally,

$$\gamma_k \leq \Lambda_k(T_0)\,. \tag{20}$$

Indeed the definition of m_k gives $\mathbb{E}[(F(C_k) - m_k); k \not\leftrightarrow Y] = 0$; splitting $\{k \not\leftrightarrow Y\}$ into $\{k \not\leftrightarrow Y \cup T_0\}$ and $\{k \not\leftrightarrow Y\} \cap \{k \leftrightarrow T_0\}$ turns this into $\gamma_k = \mathbb{E}[(F(C_k) - m_k); k \not\leftrightarrow Y, k \leftrightarrow T_0]$, and expanding the latter over the disjoint events J_x of (14) taken with k in place of o gives $\gamma_k = \Lambda_k(T_0) + \sum_{x \in T_0} \mathbb{P}(J_x)(m_x - m_k)$, in which every summand is at most zero because $r(x) < r(k)$. Now apply $\mathfrak{M}$ to (18), multiply by $P > 0$ and use (19):

$$P\,\mathfrak{M}\big[\Lambda_\cdot(T)\big] = P\,\mathfrak{M}\big[\Lambda_\cdot(T_0)\big] + \mathfrak{M}[\mathrm{Cov}^*] - \gamma_k\,\mathfrak{M}[P]\,.$$

The middle term is nonnegative. Replacing γ_k by the larger $\Lambda_k(T_0)$, which is legitimate because $\mathfrak{M}[P] \geq 0$, and writing $P(u) = P\,\pi_k(u)$ with $\pi_k(u) = \mathbb{P}(u \in C_k \mid k \not\leftrightarrow Y \cup T_0)$, the right side is at least P times the value of $\mathfrak{M}$ at $u \mapsto \Lambda_u(T_0)$ for the list obtained by inserting k in front of $z_1, \ldots, z_\ell$, by the recursion in (8) and linearity; that value is nonnegative by the induction hypothesis. Dividing by P gives (17).

We now take $\ell = 0$ in (17). The conclusion is $\Lambda_o(T) \geq \tau \Lambda_v(T)$ with $\tau = \mathbb{P}(o \in C_v \mid v \not\leftrightarrow Y \cup T)$, which is the assertion after multiplication by $\mathbb{P}(v \not\leftrightarrow Y \cup T) > 0$. If $o = v$, then the two sides of the assertion are equal. If o lies in Y or in T, then the event $\{v \not\leftrightarrow Y \cup T\} \cap \{o \leftrightarrow v\}$ is empty, so the left side vanishes, while the right side is nonnegative by the two cases settled after (16).

It remains to pass from weights strictly inside the unit interval to weights in $[0, 1]$. The right side of (16) is a finite sum, over the configurations, of the probability of the configuration, a polynomial in the pair weights, times a value depending on the weights only through the finitely many numbers m_x, each a ratio of polynomials whose denominator is positive at the given weight vector. Both sides of the asserted inequality are therefore continuous at that vector. Approximate it by weight vectors with all coordinates strictly inside the unit interval and choose for each of them an injective r ordered by its own numbers m_x; by (16) neither side depends on that choice, so the case already proved applies to each of them, and continuity gives the inequality at the given vector. $\square$

Proof of Theorem 4.1. We prove by induction on the cardinality of T that $\Lambda_o(T) \geq 0$ for every vertex o. In the base case $T = \varnothing$ the quantity (15) vanishes. For the inductive step, let T be nonempty, let $k \in T$ have the largest value of r, put $T_0 = T \setminus \{k\}$, and keep the four abbreviations P, $P(u)$, I and $I(u)$ of the proof of Proposition 4.2. The identity (18) with $u = o$ reads $\Lambda_o(T) = \Lambda_o(T_0) + I(o) - m_k P(o)$. Apply (6) with $s = k$, with $X = Y \cup T_0$ and with the two increasing functions $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$ and the indicator of the event that o lies in the cluster of k; the result is $P(o)\,I \leq P\,I(o)$, which rearranges to

$$P\big(m_k P(o) - I(o)\big) \leq P(o)\big(m_k P - I\big) = P(o)\,\gamma_k\,.$$

By (20) and $P(o) \geq 0$ the right side is at most $P(o)\Lambda_k(T_0)$, and by Proposition 4.2 with $v = k$ that is at most $P\,\Lambda_o(T_0)$. Hence

$$P\big(m_k P(o) - I(o)\big) \leq P\,\Lambda_o(T_0), \qquad \text{that is} \qquad P\,\Lambda_o(T) \geq 0\,.$$

If $P > 0$, then this is the assertion. If $P = 0$, then $P(o) = 0$ and $I(o) = 0$, so $\Lambda_o(T) = \Lambda_o(T_0)$, which is nonnegative by the induction hypothesis. $\square$

The next section turns Theorem 4.1 into a statement in which the vertex realising the minimum is fixed before the event $\{o \leftrightarrow A\}$ is imposed.

5 A Prescribed Vertex Works in Conjecture 4

The obstacle to reading Conjecture 2.4 off Theorem 4.1 is that the numbers m_x of (13) are conditional means, whereas Conjecture 2.4 compares unconditional ones. The following function removes the difference. For a vertex a, an increasing F and a set K of vertices, write $\Phi_a(K)$ for $F(K)$ minus the mean of $F(C_a)$ in a configuration in which every pair meeting K is closed and every remaining pair keeps its original weight.

Lemma 5.1. *Let a and o be vertices and let F be an increasing function of subsets of V. Then Φ_a is increasing, and:*

(i) *for every family $\mathcal{S}$ of sets of pairs,*

$$\mathbb{E}\big[F(C_o) - F(C_a);\, o \nleftrightarrow a,\ \mathcal{C}_o \in \mathcal{S}\big] = \mathbb{E}\big[\Phi_a(C_o);\, o \nleftrightarrow a,\ \mathcal{C}_o \in \mathcal{S}\big];$$

(ii) $\mathbb{E}\big[\Phi_a(C_o);\, o \nleftrightarrow a\big] = \mathbb{E}\big[F(C_o)\big] - \mathbb{E}\big[F(C_a)\big]$.

Proof. Enlarging K increases $F(K)$ and deletes more pairs, which decreases the subtracted mean, so Φ_a is increasing.

For (i), condition on $\mathcal{C}_o$. On $\{o \nleftrightarrow a\}$ the vertex a lies outside C_o, every pair with exactly one endpoint in C_o is closed, and the pairs with both endpoints outside C_o retain their original weights. Hence the conditional mean of $F(C_a)$ given $\mathcal{C}_o$ is $F(C_o) - \Phi_a(C_o)$. Both events in the statement are determined by $\mathcal{C}_o$, so summing over the values of $\mathcal{C}_o$ in $\mathcal{S}$ gives (i).

For (ii), take $\mathcal{S}$ to consist of all sets of pairs in (i). On $\{o \leftrightarrow a\}$ the two clusters coincide, so $F(C_o) - F(C_a)$ vanishes there and the left side of (i) equals $\mathbb{E}[F(C_o) - F(C_a)]$. ☐

With Φ_a in hand we can prescribe the minimising vertex before the event $\{o \leftrightarrow A\}$ is imposed, which is the form Conjecture 2.4 needs.

Theorem 5.2. *Let A be a nonempty finite set of vertices, let $a \in A$, let o be a vertex and let F be increasing with $\mathbb{E}\big[F(C_a)\big] \le \mathbb{E}\big[F(C_x)\big]$ for every $x \in A$. Then*

$$\mathbb{E}\big[F(C_a);\, o \leftrightarrow A\big] \le \mathbb{E}\big[F(C_o);\, o \leftrightarrow A\big]\,.$$

Proof. Let T be the set of $x \in A \setminus \{a\}$ with $\mathbb{P}(x \nleftrightarrow a) > 0$. Apply Theorem 4.1 with $Y = \{a\}$, with the set T, with the increasing function Φ_a of Lemma 5.1 and with the vertex o. By Lemma 5.1(ii) applied at x in place of o, the number (13) attached to $x \in T$ is

$$\frac{\mathbb{E}\big[\Phi_a(C_x);\, x \nleftrightarrow a\big]}{\mathbb{P}(x \nleftrightarrow a)} = \frac{\mathbb{E}\big[F(C_x)\big] - \mathbb{E}\big[F(C_a)\big]}{\mathbb{P}(x \nleftrightarrow a)} \ge 0\,,$$

so the sum subtracted in (15) is nonnegative and Theorem 4.1 gives

$$\mathbb{E}\big[\Phi_a(C_o);\, o \nleftrightarrow a,\ o \leftrightarrow T\big] \ge 0\,.$$

The event $\{o \leftrightarrow T\}$ is determined by $\mathcal{C}_o$, so Lemma 5.1(i) converts this into

$$\mathbb{E}\big[F(C_o) - F(C_a);\, o \nleftrightarrow a,\ o \leftrightarrow T\big] \ge 0\,. \tag{21}$$

Split $\{o \leftrightarrow A\}$ into $\{o \leftrightarrow a\}$, on which the integrand $F(C_o) - F(C_a)$ vanishes, and $\{o \nleftrightarrow a\} \cap \{o \leftrightarrow A \setminus \{a\}\}$. The latter differs from $\{o \nleftrightarrow a\} \cap \{o \leftrightarrow T\}$ only by configurations in which o misses a and reaches at least one vertex x of $A \setminus \{a\}$ outside T; such a configuration lies in the null event $\{x \nleftrightarrow a\}$, and a finite union of null events is null. Hence the left side of (21) equals $\mathbb{E}[F(C_o) - F(C_a); o \leftrightarrow A]$. ☐

Conjecture 2.4 follows at once by choosing a to realise the minimum of $\mathbb{E}\big[F(C_x)\big]$ over $x \in A$, provided every monotone cluster property is of the form $\omega \mapsto F(C_v(\omega))$ for an increasing F. That is the next lemma, whose hypotheses omit the requirement that $f(v, \omega)$ depend only on C_v, since that follows from the requirement that $f(v, \omega)$ be increasing in C_v.

Lemma 5.3. *Let f be a function of a vertex and a configuration such that $f(v,\omega) \le f(v,\xi)$ whenever $C_v(\omega) \subseteq C_v(\xi)$ and such that $f(v,\omega) = f(w,\omega)$ whenever the pair $\{v,w\}$ is open in ω. Then there is an increasing function F of vertex sets with $f(v,\omega) = F(C_v(\omega))$ for every vertex v and every configuration ω.*

Proof. For a nonempty vertex set S let ω_S be the configuration in which the open pairs are exactly those with both endpoints in S. Every vertex of S has open cluster S in ω_S, and for distinct $v,w \in S$ the pair $\{v,w\}$ is open in ω_S, so $f(v,\omega_S)$ does not depend on the choice of $v \in S$; call the common value $F(S)$. If $S \subseteq S_1$ are nonempty and $v \in S$ then $C_v(\omega_S) = S \subseteq S_1 = C_v(\omega_{S_1})$, so $F(S) \le F(S_1)$. Extend F to the empty set by a value at most $F(S)$ for every nonempty S. Finally, for a vertex v and a configuration ω put $S = C_v(\omega)$; then $C_v(\omega_S) = S = C_v(\omega)$, and the hypothesis applied in both directions gives $f(v,\omega) = f(v,\omega_S) = F(S)$. □

The two preceding results combine at once.

Corollary 5.4. *Conjecture 2.4 holds. In particular it holds for $f(v,\omega) = |C_v|$ and for $f(v,\omega) = \mathbf{1}\{|C_v| \ge k\}$ for every k.*

Proof. By Lemma 5.3 write $f(v,\cdot) = F(C_v)$ with F increasing, choose $a \in A$ realising the minimum of $\mathbb{E}\big[F(C_x)\big]$ over $x \in A$, and apply Theorem 5.2. The two displayed examples are increasing functions of the vertex set. □

The remaining conjectures of the paper follow by specialising F to an indicator.

Corollary 5.5. *The inequality* (1) *holds, and so do Conjectures 2.1, 2.2 and 2.3; Conjecture 2.3 holds with $\delta = \varepsilon/2$. Question 2.7 has an affirmative answer.*

Proof. Take $F = \mathbf{1}\{b \in \cdot\}$, so that $\mathbb{E}[F(C_x); o \leftrightarrow A] = \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ and $\mathbb{E}F(C_x) = \mathbb{P}(x \leftrightarrow b)$. Corollary 5.4 is then (1), and dropping the event $\{o \leftrightarrow A\}$ on the left gives Conjecture 2.2. If a minimises $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$ then Theorem 5.2 gives $\mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) \le \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$, which is (4), so Question 2.7 has an affirmative answer.

For Conjecture 2.1, the events $\{x \leftrightarrow b\}$ and $\{o \leftrightarrow A\}$ are both increasing, so the Harris inequality gives $\mathbb{P}(o \leftrightarrow A)\mathbb{P}(x \leftrightarrow b) \le \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$ for every $x \in A$. Taking the minimum over $x \in A$ and using (1),

$$\mathbb{P}(o \leftrightarrow A) \min_{x \in A} \mathbb{P}(x \leftrightarrow b) \le \min_{x \in A} \mathbb{P}(x \leftrightarrow b,\ o \leftrightarrow A) \le \mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \le \mathbb{P}(o \leftrightarrow b)\,.$$

For Conjecture 2.3 take $\delta = \varepsilon/2$. If $\varepsilon \ge 2$ then $1 - \varepsilon \le -1 < \mathbb{P}(o \leftrightarrow b)$ and there is nothing to prove. Otherwise $1-\varepsilon/2$ is positive, and Conjecture 2.1 gives $\mathbb{P}(o \leftrightarrow b) > (1-\varepsilon/2)^2 = 1-\varepsilon+\varepsilon^2/4 \ge 1-\varepsilon$. □

6 One Coefficient Vector for Every Target

Question 2.6 asks for one probability vector on A that works simultaneously for every vertex b. Corollary 5.4 produces, for each b, a vertex of A that works for that b; the passage from one to the other is finite-dimensional convexity. The first step is to apply Corollary 5.4 not to a single indicator but to a nonnegative weighted count of the vertices of the cluster.

Lemma 6.1. *Let A be a nonempty finite set of vertices, let o be a vertex and let $y_b \ge 0$ for $b \in V$. Then there is a vertex $a \in A$ with*

$$\sum_{b \in V} y_b\, \mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b) \le \sum_{b \in V} y_b\, \mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A)\,.$$

Proof. Put $F(K) = \sum_{b\in V} y_b \mathbf{1}\{b \in K\}$, an increasing function of vertex sets because each y_b is nonnegative. For every vertex x,

$$\mathbb{E}\big[F(C_x);\ o \leftrightarrow A\big] = \sum_{b\in V} y_b\, \mathbb{P}(x \leftrightarrow b,\ o \leftrightarrow A)\,.$$

Let $a \in A$ realise the minimum of $\mathbb{E}[F(C_x); o \leftrightarrow A]$ over $x \in A$. By Corollary 5.4, the left side of the assertion is $\mathbb{E}[F(C_a); o \leftrightarrow A]$, which is at most $\mathbb{E}[F(C_o); o \leftrightarrow A]$, and the latter is the right side. $\square$

The previous lemma gives one vertex of A for each nonnegative weighting of the targets; a separation argument turns that family of vertices into a single probability vector on A.

Theorem 6.2. *For every finite weighted graph, every nonempty finite $A \subseteq V$ and every vertex o there are numbers $c_x \geq 0$ for $x \in A$ with $\sum_{x\in A} c_x = 1$ such that*

$$\sum_{x\in A} c_x\, \mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b) \leq \mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \leq \mathbb{P}(o \leftrightarrow b) \qquad \textit{for every vertex } b\,.$$

In particular Question 2.6 has an affirmative answer.

Proof. The second inequality of the assertion is event inclusion, so only the first has to be proved. For each $x \in A$, let $M(x,\cdot)$ be the vector indexed by V whose entry at b is $\mathbb{P}(o \leftrightarrow A,\ x \leftrightarrow b)$, and let $\mathbf{w}$ be the vector indexed by V whose entry at b is $\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A)$. Let $\mathcal{N}$ be the set of all sums of a convex combination of the vectors $M(x,\cdot)$ for $x \in A$ and a vector with nonnegative entries. The set $\mathcal{N}$ is convex, and it is closed because it is the sum of a compact set and a closed set in finite dimension.

Suppose $\mathbf{w} \notin \mathcal{N}$. Strict separation of the point $\mathbf{w}$ from the closed convex set $\mathcal{N}$ gives a vector y indexed by V and a real number t with $\langle y, \mathbf{w}\rangle < t \leq \langle y, \mathbf{z}\rangle$ for every $\mathbf{z} \in \mathcal{N}$. Fix $b \in V$, let $\mathbf{u}_b$ be the vector with entry one at b and zero elsewhere, and fix a point $\mathbf{z}$ of the nonempty set $\mathcal{N}$. Then $\mathbf{z} + n\,\mathbf{u}_b$ lies in $\mathcal{N}$ for every positive integer n, so $\langle y, \mathbf{z}\rangle + n\, y_b \geq t$ for every such n, which forces $y_b \geq 0$. Lemma 6.1 therefore gives a vertex $a \in A$ with $\langle y, M(a,\cdot)\rangle \leq \langle y, \mathbf{w}\rangle$. But $M(a,\cdot)$ lies in $\mathcal{N}$, so separation gives $\langle y, M(a,\cdot)\rangle \geq t > \langle y, \mathbf{w}\rangle$, a contradiction. Hence $\mathbf{w}$ lies in $\mathcal{N}$: there are $c_x \geq 0$ with $\sum_{x\in A} c_x = 1$ and a vector with nonnegative entries whose sum with $\sum_{x\in A} c_x M(x,\cdot)$ is $\mathbf{w}$. Reading this coordinate by coordinate gives the assertion. $\square$

7 A Set of Vertices in Place of One, and Conjecture 6

Conjecture 2.5 concerns a graph in which one pair $e = \{v, w\}$ has been forced open, while the vertex a realising the minimum is prescribed in the original graph. For a finite set R of vertices write $C_R = \bigcup_{s\in R} C_s$, so that $\{R \leftrightarrow B\}$ is the event $\{C_R \cap B \neq \varnothing\}$ and $\{R \not\leftrightarrow B\}$ its complement. Nothing is assumed about the vertices of R, and the set C_R need not be connected. The following theorem is the form in which Conjecture 2.5 will be proved.

Theorem 7.1. *Let A be a nonempty finite set of vertices, let $a \in A$, let R be a nonempty finite set of vertices, and let F be an increasing function of subsets of V with $\mathbb{E}\big[F(C_a)\big] \leq \mathbb{E}\big[F(C_x)\big]$ for every $x \in A$. Then*

$$\mathbb{E}\big[F(C_R) - F(C_a);\ R \not\leftrightarrow a,\ R \leftrightarrow A \setminus \{a\}\big] \geq 0\,. \tag{22}$$

Section 7.1 deduces Conjecture 2.5 from it, and the rest of the section proves it.

7.1 Forcing the Pair Open

Two elementary facts make the passage to the measure with e forced open possible: the union of the clusters of the two endpoints of e does not see the state of e, and the mean of a function blind to that state is the same under both measures.

Lemma 7.2. *Let v and w be distinct vertices, let e be the pair $\{v,w\}$, let a be a vertex, and let ω be a configuration in which e is closed. Then $C_{\{v,w\}}$ takes the same value in ω and in the configuration obtained from ω by opening e; the same holds for the events $\{\{v,w\} \nleftrightarrow B\}$ and $\{\{v,w\} \leftrightarrow B\}$ for every set B of vertices, and, on $\{\{v,w\} \nleftrightarrow a\}$, for the cluster C_a.*

Proof. Opening e can only enlarge clusters, so $C_{\{v,w\}}$ can only grow. Conversely, a path in the enlarged configuration from v or w to a vertex u that uses e splits at its uses of e into segments of ω, each of which starts at v or at w; the last segment exhibits u in $C_v \cup C_w$ computed in ω. Hence $C_{\{v,w\}}$ is unchanged, and so are the events determined by it. If $\{v,w\} \nleftrightarrow a$ in ω, then no path of ω from a meets $\{v,w\}$, so no path from a in the enlarged configuration uses e, and C_a is unchanged as well. $\square$

The second of the two facts is a statement about the two measures rather than about clusters.

Lemma 7.3. *Let e be a pair and let φ be a real function of configurations that takes the same value on a configuration in which e is closed and on the configuration obtained from it by opening e. Then the mean of φ under $\mathbb{P}^e$ equals its mean under $\mathbb{P}$.*

Proof. Condition on the states of all the pairs other than e. Under $\mathbb{P}$ and under $\mathbb{P}^e$ those states have the same joint law, since only the weight of e differs. Given them, the hypothesis says that φ takes one and the same value whether e is open or closed, so the two conditional means of φ agree, and averaging gives the assertion. $\square$

Combining the two facts with Theorem 7.1 gives Conjecture 2.5, and in a form that does not use the hypothesis printed with it.

Theorem 7.4. *Let A be a nonempty finite set of vertices, let b be a vertex, let $a \in A$ minimise $\mathbb{P}(x \leftrightarrow b)$ over $x \in A$, and let $e = \{v,w\}$ be a pair. Then*

$$\mathbb{P}^e(a \leftrightarrow b,\ v \leftrightarrow A) \le \mathbb{P}^e(v \leftrightarrow b,\ v \leftrightarrow A)\,, \tag{23}$$

Conjecture 2.5 follows, and its hypothesis (2) holds automatically: it is the case of Conjecture 2.1 with x in place of o.

Proof. Apply Theorem 7.1 with $R = \{v,w\}$ and $F = \mathbf{1}\{b \in \cdot\}$, for which $\mathbb{E}\big[F(C_x)\big] = \mathbb{P}(x \leftrightarrow b)$, so that the hypothesis on a is exactly the minimality assumed here. By Lemmas 7.2 and 7.3 the left side of (22) has the same value under $\mathbb{P}$ and under $\mathbb{P}^e$, since the integrand and the two events defining the range of integration are unchanged by opening e on the range of integration. Hence

$$\mathbb{E}^e\big[F(C_R) - F(C_a);\ R \nleftrightarrow a,\ R \leftrightarrow A \setminus \{a\}\big] \ge 0, \qquad R = \{v,w\}\,.$$

Under $\mathbb{P}^e$ the pair e is open with probability one, so almost surely $C_v = C_w = C_R$, the event $\{R \nleftrightarrow a\}$ is $\{v \nleftrightarrow a\}$ and the event $\{R \leftrightarrow A \setminus \{a\}\}$ is $\{v \leftrightarrow A \setminus \{a\}\}$. Splitting $\{v \leftrightarrow A\}$ into $\{v \leftrightarrow a\}$, on which $C_v = C_a$ and the integrand vanishes, and $\{v \nleftrightarrow a\} \cap \{v \leftrightarrow A \setminus \{a\}\}$, the last display becomes

$$\mathbb{E}^e\big[\mathbf{1}\{b \in C_v\} - \mathbf{1}\{b \in C_a\};\ v \leftrightarrow A\big] \ge 0\,,$$

which is (23). The events $\{a \leftrightarrow b\}$ and $\{v \leftrightarrow A\}$ are increasing, so the Harris inequality and (23) give

$$\mathbb{P}^e(v \leftrightarrow A)\mathbb{P}^e(a \leftrightarrow b) \le \mathbb{P}^e(a \leftrightarrow b,\ v \leftrightarrow A) \le \mathbb{P}^e(v \leftrightarrow b,\ v \leftrightarrow A) \le \mathbb{P}^e(v \leftrightarrow b)\,,$$

which is (3). Finally, Conjecture 2.1 applied with x in place of o gives $\mathbb{P}(x \leftrightarrow b) \ge \mathbb{P}(x \leftrightarrow A)\min_{y\in A}\mathbb{P}(y \leftrightarrow b) = \mathbb{P}(x \leftrightarrow A)\mathbb{P}(a \leftrightarrow b)$, which is (2). $\square$

7.2 Events Conditioned on a Set Missing Another Set

Throughout Sections 7.2 to 7.5 fix a vertex s, a set Y of vertices with $s \notin Y$, and an increasing function g of sets of pairs, and write $D = \{s \nleftrightarrow Y\}$. For finite sets R, X and B of vertices put

$$H(R; X, B) = \{R \nleftrightarrow B\} \cap \{R \leftrightarrow X\}, \tag{24}$$

the event that C_R misses B and meets X. The event $\{R \leftrightarrow s\}$ does not imply $\{R \nleftrightarrow Y\}$ even when $\{s \nleftrightarrow Y\}$ holds, because a component of C_R other than the one containing s may meet Y, so the two conditions must be carried together; for a single vertex, however, $H(\{u\}; \{s\}, Y) = D \cap \{s \leftrightarrow u\}$, since $u \leftrightarrow s$ makes the clusters of u and s equal. Set

$$\Gamma(R) = \mathbb{P}(D)\,\mathbb{E}\big[g(\mathcal{C}_s);\ H(R; \{s\}, Y)\big] - \mathbb{E}\big[g(\mathcal{C}_s);\ D\big]\,\mathbb{P}\big(H(R; \{s\}, Y)\big), \tag{25}$$

which at $R = \{u\}$ is $\mathbb{P}(D)^2$ times the conditioned covariance appearing in Theorem 3.3. The event $H(R; \{s\}, Y)$ is contained in D: on it s lies in C_R and C_R misses Y, so $C_s \subseteq C_R$ misses Y.

The functional (8) is now applied to functions of a set argument rather than of a vertex. Fix distinct vertices $z_1, \dots, z_\ell$ and a further vertex v, all outside $\{s\} \cup Y$ and distinct from each other, put $B_1 = \{s\} \cup Y$ and $B_{j+1} = B_j \cup \{z_j\}$, and put

$$\pi_j(R) = \frac{\mathbb{P}\big(H(R; \{z_j\}, B_j)\big)}{\mathbb{P}(z_j \nleftrightarrow B_j)}, \qquad \tau(R) = \frac{\mathbb{P}\big(H(R; \{v\}, B_{\ell+1})\big)}{\mathbb{P}(v \nleftrightarrow B_{\ell+1})}. \tag{26}$$

For a real function q of finite vertex sets put $q^0 = q$, then $q^j(R) = q^{j-1}(R) - \pi_j(R) q^{j-1}(\{z_j\})$ for $1 \le j \le \ell$, and finally, for a prescribed set R_0,

$$\mathfrak{M}[q] = q^\ell(R_0) - \tau(R_0)\, q^\ell(\{v\}). \tag{27}$$

If $R_0 = \{u\}$ for a vertex u outside $B_{\ell+1}$ and distinct from v, then the numbers (26) at singletons are the constants (7) with o replaced by u, because $\mathbb{P}(H(\{y\}; \{z\}, B)) = \mathbb{P}(z \nleftrightarrow B,\ z \leftrightarrow y)$ for every y, z and B; hence (27) is (8) and $\mathfrak{M}[\Gamma] \ge 0$ is Theorem 3.3. The goal of Sections 7.2 to 7.5 is the same conclusion for a set R_0 with at least two elements.

Lemma 7.5. *There are real numbers, depending only on $z_1, \dots, z_\ell$, on v and on R_0, such that $\mathfrak{M}[q]$ is $q(R_0)$ plus the corresponding linear combination of the values $q(\{u\})$ for $u \in V$, for every real function q of finite vertex sets.*

Proof. We prove the assertion by induction on ℓ. In the base case $\ell = 0$ the combination is $-\tau(R_0)$ times $q(\{v\})$. For the inductive step, let $\ell \ge 1$; the functional (27) for the list $z_1, \dots, z_\ell$ applied to q is the functional for $z_2, \dots, z_\ell$ applied to $R \mapsto q(R) - \pi_1(R) q(\{z_1\})$. By the induction hypothesis the latter is that function at R_0 plus a combination of its values at singletons; every term produced by the subtraction is a multiple of $q(\{z_1\})$, and the coefficient of $q(R_0)$ stays one. Collecting equal vertices gives the assertion. ☐

Only the value at R_0 enters with a coefficient not attached to a singleton, and that coefficient is one; this is what makes the one-sided estimates below usable.

7.3 Exploring the Cluster of a Set

Let z be a vertex, let B be a set of vertices containing $\{s\} \cup Y$ but not z, and let Θ and Ψ be as in (9) and the paragraph containing it, built from s, Y and g. Write H_R for the event $H(R; \{z\}, B)$ of (24) and E for the event $\{z \nleftrightarrow B\}$, and assume $\mathbb{P}(E) > 0$.

Lemma 7.6. *For every finite set R of vertices, $\mathbb{E}[\Theta; H_R] = -\mathbb{E}[\Psi(C_R); H_R]$.*

Proof. Sum over the possible values of $\mathcal{C}_R$. On H_R the vertex set C_R misses B, hence misses $\{s\} \cup Y$, and every pair with exactly one endpoint in C_R is closed. Given that data, the pairs with both endpoints outside C_R retain their original weights, and deleting every pair meeting C_R changes neither $\mathcal{C}_s$ nor $\mathcal{C}_Y$ nor the quantity $\bar{g}$ of (9). The conditional mean of Θ is therefore its mean in a configuration with every pair meeting C_R deleted, which is $-\Psi(C_R)$. The event H_R is determined by the same data, so summing over its values gives the assertion. □

The gap between the value of Ψ at the cluster of R and its value at the cluster of z is what a set of vertices gains over a single vertex, and it is what the estimate below must absorb.

Lemma 7.7. *Let $\Delta(R) = \mathbb{E}[\Psi(C_R) - \Psi(C_z); H_R]$ for every finite set R of vertices. Then $\Delta(R) \geq 0$ for every such R, and $\Delta(\{u\}) = 0$ for every vertex u.*

Proof. On H_R at least one vertex of R is joined to z, so $C_z \subseteq C_R$ and $\Psi(C_z) \leq \Psi(C_R)$ because Ψ is increasing. If $R = \{u\}$ then on $H_{\{u\}}$ the clusters of u and z coincide, so the integrand vanishes identically. □

The next inequality is the step that transfers the estimate from a single vertex to a set. Let $\Gamma_{z,B}$ denote the expression (25) formed with the vertex z, the set B and the increasing function $\mathcal{C} \mapsto \Psi(\{z\} \cup \operatorname{span} \mathcal{C})$ in place of s, Y and g.

Lemma 7.8. *For every finite set R of vertices,*

$$\mathbb{E}[\Theta; H_R] - \pi(R)\, \mathbb{E}[\Theta; E] = -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)} - \Delta(R), \qquad \pi(R) = \frac{\mathbb{P}(H_R)}{\mathbb{P}(E)}, \tag{28}$$

and consequently

$$\mathbb{E}[\Theta; H_R] - \pi(R)\, \mathbb{E}[\Theta; E] \leq -\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)}, \tag{29}$$

with equality in (29) *whenever R is a single vertex.*

Proof. Expanding (25) for the vertex z, the set B and the stated function, and using that $\Psi(\{z\} \cup \operatorname{span} \mathcal{C}_z) = \Psi(C_z)$,

$$\frac{\Gamma_{z,B}(R)}{\mathbb{P}(E)} = \mathbb{E}[\Psi(C_z); H_R] - \pi(R)\, \mathbb{E}[\Psi(C_z); E].$$

By Lemma 7.6 and the definition of Δ, $\mathbb{E}[\Theta; H_R] = -\mathbb{E}[\Psi(C_z); H_R] - \Delta(R)$, and by (10), $\mathbb{E}[\Theta; E] = -\mathbb{E}[\Psi(C_z); E]$. Substituting gives (28). The inequality (29) follows from $\Delta \geq 0$, with equality at a single vertex by Lemma 7.7. □

7.4 Transfer between Two Vertex Sets in a Restricted Graph

Throughout this subsection, U is a set of vertices, $\mathbb{P}_U$, $\mathbb{E}_U$ and Cov_U are the probability, the expectation and the covariance of Section 3 taken with $W = U$, the set X is a set of vertices contained in U and containing s, the vertex v lies in U and outside X, and R is a finite set of vertices contained in U and disjoint from $X \cup \{v\}$. Every quantity formed inside U is zero if a distinguished vertex appearing in it lies outside U, and every set appearing in it is intersected with U. For example $\eta_v(W) = 0$ when $v \notin W$, as recorded in (11), and $\mathbb{P}_W(R \nleftrightarrow B)$ means $\mathbb{P}_W\big((R \cap W) \nleftrightarrow (B \cap W)\big)$. For a set W of vertices, write

$$\varrho(W) = \begin{cases} \mathbb{P}_W(R \nleftrightarrow X,\ R \leftrightarrow v)/\mathbb{P}_W(v \nleftrightarrow X), & \text{if } \mathbb{P}_W(v \nleftrightarrow X) > 0, \\ 0, & \text{if } \mathbb{P}_W(v \nleftrightarrow X) = 0. \end{cases}$$

The next three lemmas move the avoided set out of one probability and into another, so that the set R can be compared with the single vertex v. For a set K of vertices put

$$\alpha_{R,B}(K) = \mathbf{1}\{K \cap R \neq \varnothing\}\ \mathbb{P}_{U \setminus K}\big(R \setminus K \nleftrightarrow B\big). \tag{30}$$

Lemma 7.9. *The function $K \mapsto \alpha_{R,B}(K)$ is increasing, the function $B \mapsto \alpha_{R,B}(K)$ is decreasing, and if $\{v\} \cup R \cup B \subseteq U$ then*

$$\mathbb{E}_U\big[\alpha_{R,B}(C_v);\ v \nleftrightarrow B\big] = \mathbb{P}_U\big(R \nleftrightarrow B,\ R \leftrightarrow v\big)\,. \tag{31}$$

Proof. Let $K \subseteq K_1$. The first factor can only turn from zero to one. For the second, couple the two measures so that the configuration inside $U \setminus K_1$ is contained in the one inside $U \setminus K$; since also $R \setminus K_1 \subseteq R \setminus K$, avoidance of B by $R \setminus K$ inside $U \setminus K$ forces avoidance of B by $R \setminus K_1$ inside $U \setminus K_1$, so the second factor increases as well. Enlarging B makes avoidance harder and decreases the second factor. For (31), sum over the possible values of the complete set of open pairs met by paths from v. On $\{v \nleftrightarrow B\}$ the vertex set $K = C_v$ misses B, the event $\{R \leftrightarrow v\}$ is $\{K \cap R \neq \varnothing\}$, every pair with exactly one endpoint in K is closed, and the pairs with both endpoints outside K retain their original weights. On that data the event $\{R \nleftrightarrow B\}$ is the event that $R \setminus K$ misses B in $U \setminus K$, since the vertices of R inside K have cluster C_v, which misses B. $\square$

The identity (31) lets the conditional positive association of van den Berg, Häggström and Kahn be applied with the set R hidden inside an increasing function of the cluster of v.

Lemma 7.10. *Let N and N_1 be disjoint sets of vertices, and let all the sets and vertices named lie in U. Then*

$$\mathbb{P}_U\big(R \nleftrightarrow X \cup N,\ R \leftrightarrow v\big)\,\mathbb{P}_U\big(v \nleftrightarrow X \cup N_1\big) \leq \mathbb{P}_U\big(v \nleftrightarrow X \cup N \cup N_1\big)\,\mathbb{P}_U\big(R \nleftrightarrow X,\ R \leftrightarrow v\big)\,.$$

Proof. By (31) the left side is $\mathbb{E}_U[\alpha_{R,X\cup N}(C_v); v \nleftrightarrow X \cup N]\,\mathbb{P}_U(v \nleftrightarrow X \cup N_1)$. Apply Theorem 3.1 inside U with the vertex v, the two sets $X \cup N$ and $X \cup N_1$, whose intersection is X and whose union is $X \cup N \cup N_1$ because N and N_1 are disjoint, and the increasing functions $\alpha_{R,X\cup N}$ and the constant one. The result is at most $\mathbb{E}_U[\alpha_{R,X\cup N}(C_v); v \nleftrightarrow X]\,\mathbb{P}_U(v \nleftrightarrow X \cup N \cup N_1)$, whose first factor is at most $\mathbb{E}_U[\alpha_{R,X}(C_v); v \nleftrightarrow X] = \mathbb{P}_U(R \nleftrightarrow X, R \leftrightarrow v)$ by Lemma 7.9 and (31). If v lies in $X \cup N$ or R meets $X \cup N$, then the left side is zero. $\square$

The previous lemma handles two disjoint avoided sets; the next removes that restriction by exposing the overlap and applying the four functions theorem.

Lemma 7.11. *Let U be a set of vertices, let $X \subseteq U$ contain s, let $v \in U \setminus X$, let $R \subseteq U$ be disjoint from $X \cup \{v\}$, and let N and N_1 be subsets of U. Let η be a nonnegative real function of subsets of U that vanishes on sets not containing v, and let*

$$\phi(N) = \mathbb{E}_U\big[\mathbf{1}\{s \nleftrightarrow N\}\,\eta(U \setminus C_N)\big], \qquad \chi(N) = \mathbb{E}_U\big[\mathbf{1}\{s \nleftrightarrow N\}\mathbf{1}\{R \nleftrightarrow N\}\,\varrho(U \setminus C_N)\,\eta(U \setminus C_N)\big]\,.$$

Assume that for every $W \subseteq U$ and every $Q \subseteq W$ the function ϕ formed inside W satisfies $\phi(Q)\mathbb{P}_W(v \nleftrightarrow X) \leq \mathbb{P}_W(v \nleftrightarrow X \cup Q)\,\eta(W)$, and that $Q \mapsto \phi(Q)$ is decreasing in every such W. Then

$$\mathbb{P}_U\big(R \nleftrightarrow X \cup N,\ R \leftrightarrow v\big)\,\phi(N_1) \leq \mathbb{P}_U\big(v \nleftrightarrow X \cup N \cup N_1\big)\,\chi(N \cap N_1)\,. \tag{32}$$

Proof. We prove (32) by induction on the number of elements of U. In the base case $U = \varnothing$ every probability in (32) is that of a configuration with no pairs, and both sides vanish. Suppose first that N and N_1 are disjoint. Multiply the assumed inequality for N_1 by $\mathbb{P}_U(R \nleftrightarrow X \cup N,\ R \leftrightarrow v)$ and apply Lemma 7.10, using $\mathbb{P}_U(R \nleftrightarrow X, R \leftrightarrow v) = \varrho(U)\mathbb{P}_U(v \nleftrightarrow X)$ and $\chi(\varnothing) = \varrho(U)\eta(U)$. If $\mathbb{P}_U(v \nleftrightarrow X) = 0$, then $\mathbb{P}_U(R \nleftrightarrow X \cup N,\ R \leftrightarrow v) = 0$ by event inclusion and both sides vanish.

Otherwise put $Z = N \cap N_1$, which is nonempty. If Z contains v or s, or if R meets Z, then the left side is zero. In the remaining case remove Z and expose the set $\Sigma \subseteq U \setminus Z$ of vertices joined to Z by an open pair. The indicators of the events $\{u \in \Sigma\}$ are determined by disjoint blocks of pairs, so the law of Σ is a product measure, written $\mathbb{P}_\Sigma$, on subsets of $U \setminus Z$. Put $N_2 = N \setminus Z$ and $N_3 = N_1 \setminus Z$, let ϕ and χ be formed inside $U \setminus Z$, and put

$$\kappa(\Sigma) = \mathbb{P}_{U\setminus Z}\big(R \nleftrightarrow X \cup N_2 \cup \Sigma,\ R \leftrightarrow v\big), \qquad \mu(\Sigma) = \mathbb{P}_{U\setminus Z}\big(v \nleftrightarrow X \cup N_2 \cup N_3 \cup \Sigma\big)\,.$$

The four quantities in (32) are the $\mathbb{P}_\Sigma$-averages of $\kappa(\Sigma)$, of $\phi(N_3\cup\Sigma)$, of $\mu(\Sigma)$ and of $\chi(\Sigma)$: the event that R misses N in U is the event that R misses $N_2\cup\Sigma$ in $U\setminus Z$, and likewise for the other three, so no factor is taken outside an expectation. For subsets Σ_1 and Σ_2 of $U\setminus Z$, the two sets $\Sigma_1\cup(N_2\setminus\Sigma_2)$ and $\Sigma_2\cup(N_3\setminus\Sigma_1)$ have union $N_2\cup N_3\cup\Sigma_1\cup\Sigma_2$ and intersection $\Sigma_1\cap\Sigma_2$, so the induction hypothesis inside $U\setminus Z$, together with the assumed monotonicity of ϕ, gives $\kappa(\Sigma_1)\phi(N_3\cup\Sigma_2)\le\mu(\Sigma_1\cup\Sigma_2)\chi(\Sigma_1\cap\Sigma_2)$. Multiplying by the product identity $\mathbb{P}_\Sigma(\Sigma_1)\mathbb{P}_\Sigma(\Sigma_2)=\mathbb{P}_\Sigma(\Sigma_1\cap\Sigma_2)\mathbb{P}_\Sigma(\Sigma_1\cup\Sigma_2)$ and applying the Ahlswede and Daykin four functions theorem to the four functions $\Sigma\mapsto\mathbb{P}_\Sigma(\Sigma)\kappa(\Sigma)$, $\Sigma\mapsto\mathbb{P}_\Sigma(\Sigma)\phi(N_3\cup\Sigma)$, $\Sigma\mapsto\mathbb{P}_\Sigma(\Sigma)\mu(\Sigma)$ and $\Sigma\mapsto\mathbb{P}_\Sigma(\Sigma)\chi(\Sigma)$ yields (32). □

It remains to compare the covariance produced by a set of vertices with the covariance produced by a single vertex, and this is where the second inequality of van den Berg, Häggström and Kahn enters.

Lemma 7.12. *Let U be a set of vertices, let $X\subseteq U$ contain s, let $v\in U\setminus X$, let $R\subseteq U$ be disjoint from $X\cup\{v\}$, and let g be an increasing function of sets of pairs. Let $\eta_R(U)=\operatorname{Cov}_U\big(g(\mathcal{C}_s),\mathbf{1}\{R\leftrightarrow X\}\big)$ and $\eta_v(U)=\operatorname{Cov}_U\big(g(\mathcal{C}_s),\mathbf{1}\{v\leftrightarrow X\}\big)$. Then $\eta_v(U)\ge 0$ and*

$$\varrho(U)\,\eta_v(U)\le\eta_R(U)\,.$$

Proof. All expectations are taken under $\mathbb{P}_U$, and the argument U is suppressed. If $v\notin U$, then $\eta_v=0$ and the assertion is the Harris inequality. For every real number t the events $\{v\leftrightarrow X\}$ and $\{g(\mathcal{C}_s)\ge t\}$ are increasing, so $\eta_v\ge 0$ by the Harris inequality. Write J for the event $\{R\nleftrightarrow X\}\cap\{R\leftrightarrow v\}$. If $v\leftrightarrow X$ and $R\leftrightarrow v$, then $R\leftrightarrow X$, so J is contained in $\{v\nleftrightarrow X\}$, and $\mathbf{1}\{R\leftrightarrow X\}+\mathbf{1}_J=\mathbf{1}\{R\leftrightarrow X \text{ or } R\leftrightarrow v\}$, an increasing event. The Harris inequality applied to that event gives $\eta_R+\operatorname{Cov}_U(g(\mathcal{C}_s),\mathbf{1}_J)\ge 0$, that is

$$\eta_R\ge\mathbb{E}_U[g(\mathcal{C}_s)]\,\mathbb{P}_U(J)-\mathbb{E}_U[g(\mathcal{C}_s);J]\,.$$

Write $\mathcal{C}_X$ and $\mathcal{C}_v$ for the sets of open pairs lying on open paths from X and from v. The set $\mathcal{C}_s$ is the connected component of $\mathcal{C}_X$ containing s, so $g(\mathcal{C}_s)$ is an increasing function of $\mathcal{C}_X$ and does not depend on $\mathcal{C}_v$; the event $\{R\nleftrightarrow X\}$ is $\{R\cap(X\cup\operatorname{span}\mathcal{C}_X)=\varnothing\}$ and the event $\{R\leftrightarrow v\}$ is $\{R\cap(\{v\}\cup\operatorname{span}\mathcal{C}_v)\ne\varnothing\}$, so $\mathbf{1}_J$ is decreasing in $\mathcal{C}_X$ and increasing in $\mathcal{C}_v$. Hence (5), applied with $S_1=X$, $S_2=\{v\}$ and the two functions $g(\mathcal{C}_s)$ and $\mathbf{1}_J$, gives $\mathbb{P}_U(v\nleftrightarrow X)\mathbb{E}_U[g(\mathcal{C}_s);J]\le\mathbb{P}_U(J)\mathbb{E}_U[g(\mathcal{C}_s);v\nleftrightarrow X]$. Since $\eta_v=\mathbb{E}_U[g(\mathcal{C}_s)]\mathbb{P}_U(v\nleftrightarrow X)-\mathbb{E}_U[g(\mathcal{C}_s);v\nleftrightarrow X]$ and, when $\mathbb{P}_U(v\nleftrightarrow X)>0$, $\varrho(U)=\mathbb{P}_U(J)/\mathbb{P}_U(v\nleftrightarrow X)$,

$$\varrho(U)\eta_v=\mathbb{E}_U[g(\mathcal{C}_s)]\mathbb{P}_U(J)-\frac{\mathbb{P}_U(J)\,\mathbb{E}_U[g(\mathcal{C}_s);v\nleftrightarrow X]}{\mathbb{P}_U(v\nleftrightarrow X)}\le\mathbb{E}_U[g(\mathcal{C}_s)]\mathbb{P}_U(J)-\mathbb{E}_U[g(\mathcal{C}_s);J]\le\eta_R\,.$$

If $\mathbb{P}_U(v\nleftrightarrow X)=0$, then $\varrho(U)=0$ and $\eta_v=0$, and the assertion is the Harris inequality. □

The two preceding lemmas combine into the inequality that replaces the single vertex v by the set R_0 once the cluster of Y has been exposed. Write $U=V\setminus C_Y$ and

$$\mathcal{H}\coloneqq\mathbb{E}\big[\mathbf{1}_D\mathbf{1}\{R_0\nleftrightarrow Y\}\,\eta_{R_0}(U)\big]-\tau(R_0)\,\mathbb{E}\big[\mathbf{1}_D\,\eta_v(U)\big]\,,$$

with η_{R_0} and η_v the two covariances of Lemma 7.12.

Proposition 7.13. *Let every pair weight lie strictly between zero and one, let $X=\{s\}\cup\{z_1,\dots,z_\ell\}$, let v be a vertex outside $X\cup Y$, let R_0 be a finite set of vertices disjoint from $X\cup Y\cup\{v\}$, and let g be an increasing function of sets of pairs. Then $\mathcal{H}\ge 0$.*

Proof. Subtracting from g its minimum over the finitely many values of $\mathcal{C}_s$ changes neither η_{R_0} nor η_v, so assume $g\ge 0$. Take $h=\eta_v$, which is nonnegative by Lemma 7.12 and vanishes on sets not containing v; the two hypotheses of Lemma 7.11 on h are (12) and the monotonicity recorded with it. Apply Lemma 7.11 with $U=V$, with $R=R_0$ and with $N=N_1=Y$, whose intersection and union are both Y. Its left side is $\mathbb{P}(R_0\nleftrightarrow X\cup Y,R_0\leftrightarrow v)\,\mathbb{E}[\mathbf{1}_D\eta_v(U)]$ and its right side is $\mathbb{P}(v\nleftrightarrow X\cup Y)\,\mathbb{E}[\mathbf{1}_D\mathbf{1}\{R_0\nleftrightarrow Y\}\varrho(U)\eta_v(U)]$. Dividing by $\mathbb{P}(v\nleftrightarrow X\cup Y)$, which is positive because the configuration with no open pair has positive probability, and recognising $\tau(R_0)$ as the resulting quotient by (26) with $B_{\ell+1} = X \cup Y$,

$$\tau(R_0)\,\mathbb{E}\big[\mathbf{1}_D\,\eta_v(U)\big] \le \mathbb{E}\big[\mathbf{1}_D\mathbf{1}\{R_0 \nleftrightarrow Y\}\,\varrho(U)\,\eta_v(U)\big]\,.$$

By Lemma 7.12 applied inside U, the integrand on the right is at most $\mathbf{1}_D\mathbf{1}\{R_0 \nleftrightarrow Y\}\eta_{R_0}(U)$. $\square$

7.5 The Inequality for a Set of Vertices

Recall Θ from (9) and put

$$\mathcal{W} = \mathbb{E}\big[\Theta \cdot \mathfrak{M}[\beta]\big]\,, \tag{33}$$

where $\beta(R) = \mathbf{1}\{R \cap C_Y = \varnothing\}\mathbf{1}\{R \leftrightarrow s\}$ is the indicator of $H(R; \{s\}, Y)$ and $\mathfrak{M}$ acts on the argument R with the values of $\mathcal{C}_s$ and $\mathcal{C}_Y$ held fixed. Because $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$, the quantity $\mathcal{W}$ is the average over $\mathcal{C}_Y$ of a conditional covariance, and it is the part of $\mathfrak{M}[\Gamma]$ produced after the cluster of Y has been exposed.

For $1 \le j \le \ell$, let $\mathfrak{M}_j$ be the functional (27) built from $z_{j+1}, \dots, z_\ell$, from v and from R_0 with base set B_{j+1}.

Proposition 7.14. *Let every pair weight lie strictly between zero and one, let $z_1, \dots, z_\ell$ be distinct vertices outside $\{s\} \cup Y$, let v be a further vertex outside $\{s\} \cup Y \cup \{z_1, \dots, z_\ell\}$, let R_0 be a finite set of vertices disjoint from $\{s, v\} \cup Y \cup \{z_1, \dots, z_\ell\}$, and let g be an increasing function of sets of pairs. Then*

$$\mathcal{W} = \mathcal{H} + \sum_{j=1}^{\ell} \frac{\mathfrak{M}_j\left[\Gamma_{z_j,B_j}\right]}{\mathbb{P}(z_j \nleftrightarrow B_j)} + \sum_{j=1}^{\ell} \Delta_j(R_0)\,, \tag{34}$$

where Γ_{z_j,B_j} and Δ_j are the quantities of Section 7.3 for the vertex z_j and the set B_j.

Proof. The idea is to peel the vertices $z_1, \dots, z_\ell$ off the test function β one at a time, each peeling producing one term of (34), and to identify what is left after ℓ peelings with $\mathcal{H}$.

Write $Z = C_Y$ and, for a set W of vertices and a vertex z,

$$\beta_W(R) = \mathbf{1}\{R \cap Z = \varnothing,\ R \leftrightarrow W\}, \qquad \delta_{W,z}(R) = \mathbf{1}\{R \cap Z = \varnothing,\ R \nleftrightarrow W,\ R \leftrightarrow z\}\,.$$

Two pointwise identities hold: $\beta_{W \cup \{z\}} = \beta_W + \delta_{W,z}$, and $\beta_W(\{z\}) = \mathbf{1}\{z \notin Z\} - \delta_{W,z}(\{z\})$, the latter because connection is reflexive. Hence, for the operator $q \mapsto q(\cdot) - \pi(\cdot)q(\{z\})$ with π as in (26),

$$\beta_W(R) - \pi(R)\beta_W(\{z\}) = \beta_{W \cup \{z\}}(R) - \big(\delta_{W,z}(R) - \pi(R)\delta_{W,z}(\{z\})\big) - \pi(R)\mathbf{1}\{z \notin Z\}\,.$$

Multiply by Θ and take expectations. Every later operator and the final subtraction have constant coefficients, so the last term contributes a constant multiple of $\mathbb{E}[\Theta\mathbf{1}\{z \notin Z\}]$, which vanishes because $\mathbf{1}\{z \notin Z\}$ is a function of $\mathcal{C}_Y$ and $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$. With $W = \{s\} \cup \{z_1, \dots, z_{j-1}\}$ and $z = z_j$, the event $\{R \cap Z = \varnothing, R \nleftrightarrow W\}$ is $\{R \nleftrightarrow B_j\}$, and $\delta_{W,z_j}(\{z_j\})$ is the indicator of $\{z_j \nleftrightarrow B_j\}$; the middle bracket is therefore the left side of (28) for the vertex z_j and the set B_j. Applying (28),

$$\mathbb{E}\big[\Theta\big(\beta_W(R) - \pi(R)\beta_W(\{z_j\})\big)\big] = \mathbb{E}\big[\Theta\,\beta_{W \cup \{z_j\}}(R)\big] + \frac{\Gamma_{z_j,B_j}(R)}{\mathbb{P}(z_j \nleftrightarrow B_j)} + \Delta_j(R)\,.$$

Start from $\beta_{\{s\}}$, which is the β of (33), iterate from $j = 1$ to $j = \ell$ and then subtract $\tau(R_0)$ times the value at $\{v\}$. The terms produced at step j are acted on only by the operators of the vertices after z_j and by the final subtraction, which together form $\mathfrak{M}_j$; by Lemma 7.5 applied to $\mathfrak{M}_j$ they are evaluated at R_0 with coefficient one and at singletons otherwise, and Δ_j vanishes at every singleton by Lemma 7.7, so only $\Delta_j(R_0)$ survives among those terms. After all ℓ steps the remaining test function is β_X with $X = \{s\} \cup \{z_1, \dots, z_\ell\}$, and

$$\mathbb{E}\big[\Theta\big(\beta_X(R_0) - \tau(R_0)\beta_X(\{v\})\big)\big] = \mathcal{H}\,,$$

because conditioning on $\mathcal{C}_Y$ and using $\mathbb{E}[\Theta \mid \mathcal{C}_Y] = 0$ turns each of the two terms into the conditional covariance of $g(\mathcal{C}_s)$ with the corresponding connection indicator inside $U = V \setminus C_Y$, which are $\eta_{R_0}(U)$

and $\eta_v(U)$ of Lemma 7.12, carrying the factors $\mathbf{1}_D \mathbf{1}\{R_0 \nleftrightarrow Y\}$ and $\mathbf{1}_D$, the latter because $\eta_v(U)$ vanishes when $v \notin U$. This is (34). $\square$

The expansion (34) now yields the goal of this subsection by an induction in which each term has already been shown to be nonnegative.

Proposition 7.15. *Let every pair weight lie strictly between zero and one, let Y be a set of vertices, let s be a vertex outside Y, let $z_1, \ldots, z_\ell$ be distinct vertices outside $\{s\} \cup Y$, let v be a further vertex outside $\{s\} \cup Y \cup \{z_1, \ldots, z_\ell\}$, and let R_0 be a finite set of vertices with at least two elements, disjoint from $\{s, v\} \cup Y \cup \{z_1, \ldots, z_\ell\}$. Then $\mathfrak{M}[\Gamma] \geq 0$ for every increasing function g of sets of pairs.*

Proof. We prove by induction on ℓ, for all admissible data at once, both that $\mathcal{W} \geq 0$ for every increasing g and that $\mathfrak{M}[\Gamma] \geq 0$ for every increasing g. The induction proves two things at each ℓ, and we take them as two steps.

For the first step we show that $\mathcal{W} \geq 0$. Replace g by g minus its minimum, apply Proposition 7.14 and estimate the three groups of terms in (34). The first is nonnegative by Proposition 7.13. Each denominator $\mathbb{P}(z_j \nleftrightarrow B_j)$ is positive because $z_j \notin B_j$ and the configuration with no open pair has positive probability, each $\mathfrak{M}_j[\Gamma_{z_j, B_j}]$ is a quantity of the same shape with a strictly shorter list and with the nonnegative increasing Ψ in place of g, hence nonnegative by the induction hypothesis, and each $\Delta_j(R_0)$ is nonnegative by Lemma 7.7. Hence $\mathcal{W} \geq 0$.

For the second step we expand Γ. By (25) and the containment $H(R; \{s\}, Y) \subseteq D$, and writing the two indicators of $H(R; \{s\}, Y)$ in terms of the vertex sets $\{s\} \cup \operatorname{span} \mathcal{C}_s$ and $Y \cup \operatorname{span} \mathcal{C}_Y$,

$$\Gamma(R) = \mathbb{P}(D)\, \mathbb{E}\big[g(\mathcal{C}_s)\, \beta_*(R);\, D\big] - \mathbb{E}\big[g(\mathcal{C}_s); D\big]\, \mathbb{E}\big[\beta_*(R); D\big]\,,$$

in which $\beta_*(R)$ is the indicator that R meets $\{s\} \cup \operatorname{span} \mathcal{C}_s$ and misses $Y \cup \operatorname{span} \mathcal{C}_Y$. Since $\mathfrak{M}$ is linear in its argument and its coefficients are constants, $\mathfrak{M}[\Gamma]$ is the same expression with $\beta_*(R)$ replaced by $\mathfrak{M}[\beta_*]$, a function of the two sets $\mathcal{C}_s$ and $\mathcal{C}_Y$. Modify $\mathfrak{M}[\beta_*]$ off the event D by dropping the second indicator at every singleton evaluation; on D the two clusters are disjoint and nothing changes, while the modified function is decreasing in $\mathcal{C}_Y$ because by Lemma 7.5 only the evaluation at R_0 depends on $\mathcal{C}_Y$ and it enters with coefficient one. Here the hypothesis that R_0 has at least two elements is used, since R_0 must not itself be a singleton.

For real functions q_1 and q_2 of configurations write $\operatorname{Cov}_D(q_1, q_2) = \mathbb{E}[q_1 q_2; D] - \mathbb{E}[q_1; D]\mathbb{E}[q_2; D]/\mathbb{P}(D)$, so that the last paragraph reads $\mathfrak{M}[\Gamma] = \mathbb{P}(D) \operatorname{Cov}_D(g(\mathcal{C}_s), \psi)$ with ψ the modified function of $(\mathcal{C}_s, \mathcal{C}_Y)$. Let ψ_0 be a function of $\mathcal{C}_s$ and put $\bar{\psi}_0 = \mathbb{E}[\psi_0 \mid \mathcal{C}_Y]$. Since D is determined by $\mathcal{C}_Y$ and $\mathbb{E}[\bar{\psi}_0; D] = \mathbb{E}[\psi_0; D]$,

$$\operatorname{Cov}_D(\psi_0, \psi) = \mathbb{E}\big[\mathbf{1}_D(\psi_0 - \bar{\psi}_0)\psi\big] + \mathbb{E}\big[\mathbf{1}_D(\bar{\psi}_0 - \mathbb{E}[\bar{\psi}_0 \mid \mathcal{C}_s])\psi\big] + \operatorname{Cov}_D\big(\mathbb{E}[\bar{\psi}_0 \mid \mathcal{C}_s], \psi\big)\,.$$

For $\psi_0 = g(\mathcal{C}_s)$ the first term is $\mathcal{W}$, nonnegative for every increasing g by the first half of the proof. The second term is $\mathbb{E}[\mathbf{1}_D \operatorname{Cov}(\bar{\psi}_0, \psi \mid \mathcal{C}_s)]$; given $\mathcal{C}_s$ and on D the configuration outside C_s retains its original weights, and both $\bar{\psi}_0$ and ψ are decreasing in $\mathcal{C}_Y$, hence in that configuration, so the conditional covariance is nonnegative by the Harris inequality. The third term has the same shape with ψ_0 replaced by $\mathbb{E}[\bar{\psi}_0 \mid \mathcal{C}_s]$, again an increasing function of $\mathcal{C}_s$. Iterate this two-step averaging operator $q \mapsto \mathbb{E}[\mathbb{E}[q \mid \mathcal{C}_Y] \mid \mathcal{C}_s]$ and write ψ_n for its nth iterate applied to ψ_0, so that the third term becomes $\operatorname{Cov}_D(\psi_n, \psi)$. Let $\mathcal{A}$ be the finite set of values K of $\mathcal{C}_s$ with $\mathbb{P}(\mathcal{C}_s = K,\ D) > 0$. The event D is determined both by $\mathcal{C}_s$ and by $\mathcal{C}_Y$, so the two-step operator preserves $\mathcal{A}$, and its restriction to $\mathcal{A}$ has every entry positive. Indeed, for any two values K and K_1 in $\mathcal{A}$ the value $\mathcal{C}_Y = \varnothing$ has positive joint probability with each of them: close every pair incident to Y in a configuration realising the prescribed cluster of s; this does not change a cluster of s that misses Y, and every configuration has positive probability. The term $\mathcal{C}_Y = \varnothing$ therefore makes the transition from K to K_1 positive. A finite stochastic matrix with all entries positive contracts oscillation uniformly, so the oscillation of ψ_n on $\mathcal{A}$ tends to zero, that is, ψ_n converges to a constant on D. Since Cov_D vanishes on constants and is a bounded function of its first argument, $\operatorname{Cov}_D(\psi_n, \psi) \to 0$. Hence $\mathfrak{M}[\Gamma] \geq 0$. $\square$

7.6 Removing the Vertices of the Set One at a Time

Fix a set Y, an increasing function F of vertex sets, a finite $T \subseteq V \setminus Y$ with $\mathbb{P}(x \nleftrightarrow Y) > 0$ for $x \in T$, and an injective r on T with $m_x \le m_y$ whenever $r(x) < r(y)$, the numbers m_x being those of (13). For a finite set R of vertices define, in analogy with (14) and (15),

$$J_x(R) = \{R \nleftrightarrow Y\} \cap \{R \leftrightarrow x\} \cap \bigcap_{y \in T,\ r(y) < r(x)} \{R \nleftrightarrow y\}\,,$$

$$\Lambda_R(T) = \mathbb{E}\big[F(C_R);\, R \nleftrightarrow Y,\ R \leftrightarrow T\big] - \sum_{x \in T} \mathbb{P}\big(J_x(R)\big)\, m_x\,.$$

For $R = \{u\}$ these are (14) and (15).

Theorem 7.16. *Let V be a finite vertex set with pair weights in $[0,1]$, let $Y \subseteq V$, let F be an increasing function of subsets of V, let $T \subseteq V \setminus Y$ satisfy $\mathbb{P}(x \nleftrightarrow Y) > 0$ for every $x \in T$, let r be injective on T with $m_x \le m_y$ whenever $r(x) < r(y)$, and let R be a nonempty finite set of vertices disjoint from $Y \cup T$. Then $\Lambda_R(T) \ge 0$.*

Proof. The idea of the proof is to record three facts about removing the vertex of T at which r is largest, to prove the resulting inequality for the functional $\mathfrak{M}$ by induction, and then to run the same removal argument again with $\mathfrak{M}$ specialised to a single transfer. We divide the proof into four steps: the three preliminary facts, the inequality (38) for $\mathfrak{M}$, the induction proving $\Lambda_R(T) \ge 0$, and the passage to weights in $[0,1]$.

Assume first that every pair weight lies strictly between zero and one. If R is a single vertex, then the assertion is Theorem 4.1, so assume that R has at least two elements.

For the first step, let $k \notin T$ satisfy $r(y) < r(k)$ for every $y \in T$, and write H_R for the event $H(R; \{k\}, Y \cup T)$ of (24). Adding k leaves the events $J_x(R)$ for $x \in T$ unchanged and creates the single new event H_R, so

$$\Lambda_R(T \cup \{k\}) - \Lambda_R(T) = \mathbb{E}\big[F(C_R) - m_k;\, H_R\big]\,, \tag{35}$$

and on H_R at least one vertex of R is joined to k, so $C_k \subseteq C_R$ and the right side is at least $\mathbb{E}[F(C_k) - m_k; H_R]$, with equality when R is a single vertex. Second, with $\gamma_k = m_k \mathbb{P}(k \nleftrightarrow Y \cup T) - \mathbb{E}[F(C_k);\, k \nleftrightarrow Y \cup T]$ and with $\Gamma_{k,Y\cup T}$ the quantity (25) formed with the vertex k, the set $Y \cup T$ and the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$ in the roles of s, Y and g,

$$\mathbb{P}(k \nleftrightarrow Y \cup T)\, \mathbb{E}\big[F(C_k) - m_k; H_R\big] = \Gamma_{k,Y\cup T}(R) - \gamma_k\, \mathbb{P}(H_R)\,, \tag{36}$$

as one sees by expanding both sides. Third, exactly as in the proof of Proposition 4.2,

$$\gamma_k \le \Lambda_{\{k\}}(T)\,. \tag{37}$$

Let now $\mathfrak{M}$ be the functional (27) with $R_0 = R$, with the vertices $z_1, \dots, z_\ell$ and v outside $Y \cup T \cup R$, and with the sets (26) based at $Y \cup T$. For the second step we claim that

$$\mathfrak{M}\big[\, R_1 \mapsto \Lambda_{R_1}(T)\,\big] \ge 0 \tag{38}$$

for every such choice, and we prove it by induction on the cardinality of T. In the base case $T = \varnothing$ every value vanishes. For the inductive step, let $k \in T$ have the largest value of r, put $T_0 = T \setminus \{k\}$, and read (35) for T_0 and k. By Lemma 7.5 the value at R enters $\mathfrak{M}$ with coefficient one and every other evaluation is at a singleton, where (35) is an equality; hence (35) may be inserted inside $\mathfrak{M}$ with the correct sign. Write P for $\mathbb{P}(k \nleftrightarrow Y \cup T_0)$. Multiplying by $P > 0$ and using (36),

$$P\, \mathfrak{M}\big[\Lambda.(T)\big] \ge P\, \mathfrak{M}\big[\Lambda.(T_0)\big] + \mathfrak{M}\big[\Gamma_{k,Y\cup T_0}\big] - \gamma_k\, P\, \mathfrak{M}\big[\pi\big]\,,$$

where $\pi(R_1) = \mathbb{P}(H(R_1; \{k\}, Y \cup T_0))/P$. The middle term is nonnegative by Proposition 7.15 applied to the vertex k, the set $Y \cup T_0$ and the increasing function $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span} \mathcal{C})$. The value of $\mathfrak{M}$ at

π is nonnegative: Proposition 7.15 applied to the same vertex and set and to the increasing function $\mathcal{C} \mapsto \mathbf{1}\{\mathcal{C} \neq \varnothing\}$ gives $\mathfrak{M}[\Gamma^*_{k,Y\cup T_0}] \geq 0$ for the corresponding quantity Γ^*, and a direct expansion gives $\Gamma^*(R_1) = \mathbb{P}(k \nleftrightarrow Y \cup T_0,\ \mathcal{C}_k = \varnothing)\,\mathbb{P}(H(R_1;\{k\}, Y \cup T_0))$. Indeed $\mathfrak{M}$ evaluates its argument at R and at the singletons $\{z_1\}, \dots, \{z_\ell\}, \{v\}$, none of which contains k, so on $H(R_1;\{k\}, Y \cup T_0)$ the vertex k is joined to a vertex other than itself and therefore lies in an open pair. The first factor is positive because the configuration with no open pair has positive probability, so dividing by it and by P gives $\mathfrak{M}[\pi] \geq 0$. By (37) and $\mathfrak{M}[\pi] \geq 0$,

$$\mathfrak{M}\big[\Lambda_\cdot(T_0)\big] - \gamma_k\,\mathfrak{M}[\pi] \geq \mathfrak{M}\big[\Lambda_\cdot(T_0)\big] - \Lambda_{\{k\}}(T_0)\,\mathfrak{M}[\pi]\,,$$

and by the recursion defining (27) the right side is the corresponding quantity for the set T_0 and the list obtained by prepending k to $z_1, \dots, z_\ell$, which is nonnegative by the induction hypothesis. Dividing by P gives (38).

The second step is complete. Taking the set T_0 in place of T, $\ell = 0$ and $v = k$ in (38) gives

$$\mathbb{P}\big(H(R;\{k\}, Y \cup T_0)\big)\,\Lambda_{\{k\}}(T_0) \leq \mathbb{P}\big(k \nleftrightarrow Y \cup T_0\big)\,\Lambda_R(T_0) \tag{39}$$

for every k outside $Y \cup T_0 \cup R$. For the third step we run a second induction, again on the cardinality of T, which may use (38) for every T because that inequality has just been proved for all T at once. In the base case $T = \varnothing$ the quantity vanishes. For the inductive step, let $k \in T$ have the largest value of r, put $T_0 = T \setminus \{k\}$, write P for $\mathbb{P}(k \nleftrightarrow Y \cup T_0)$ and H_R for $H(R;\{k\}, Y \cup T_0)$, so that (35) reads $\Lambda_R(T) = \Lambda_R(T_0) + \mathbb{E}[F(C_R) - m_k; H_R]$. On H_R the cluster C_k is contained in C_R, so

$$\mathbb{E}\big[F(C_R) - m_k; H_R\big] \geq \mathbb{E}\big[F(C_k) - m_k; H_R\big]\,.$$

Conditionally on $\mathcal{C}_k$ and on $\{k \nleftrightarrow Y \cup T_0\}$, the conditional probability of H_R is $\alpha_{R,Y\cup T_0}(C_k)$ with α as in (30) and $U = V$, an increasing function of that data by Lemma 7.9, and $\mathbb{E}[\alpha_{R,Y\cup T_0}(C_k);\ k \nleftrightarrow Y \cup T_0] = \mathbb{P}(H_R)$ by (31). Hence (6) applied with $s = k$, with the set $Y \cup T_0$ and with the two increasing functions $\mathcal{C} \mapsto F(\{k\} \cup \operatorname{span}\mathcal{C})$ and $\mathcal{C} \mapsto \alpha_{R,Y\cup T_0}(\{k\} \cup \operatorname{span}\mathcal{C})$ gives $P\,\mathbb{E}[F(C_k); H_R] \geq \mathbb{P}(H_R)\mathbb{E}[F(C_k);\ k \nleftrightarrow Y \cup T_0]$, whence

$$P\,\mathbb{E}\big[F(C_k) - m_k; H_R\big] \geq -\mathbb{P}(H_R)\,\gamma_k\,.$$

Multiplying (35) by P and using (39), (37) and the last display,

$$P\,\Lambda_R(T) \geq \mathbb{P}(H_R)\,\Lambda_{\{k\}}(T_0) - \mathbb{P}(H_R)\gamma_k \geq 0\,,$$

and dividing by $P > 0$ finishes the second induction.

For the fourth step it remains to pass to weights in $[0,1]$. As in (16), the assumption on r gives

$$\Lambda_R(T) = \mathbb{E}\Big[\mathbf{1}\{R \nleftrightarrow Y,\ R \leftrightarrow T\}\Big(F(C_R) - \min_{x\in T\cap C_R} m_x\Big)\Big]\,,$$

an expression not referring to r and, by the argument given after (16), continuous at every weight vector at which $\mathbb{P}(x \nleftrightarrow Y) > 0$ for each $x \in T$. Approximate such a vector by vectors with all coordinates strictly inside the unit interval and choose for each an injective r ordered by its own numbers m_x; the case already proved applies to each of them, and continuity gives the assertion. □

7.7 Proof of the Comparison for a Set of Vertices

Two identities relate F to the function Φ_a of Section 5 when a set of vertices replaces a single one.

Lemma 7.17. *Let R be a nonempty finite set of vertices, let a be a vertex, let F be increasing and let $\mathcal{S}$ be a family of sets of pairs. Then*

$$\mathbb{E}\big[F(C_R) - F(C_a);\ R \nleftrightarrow a,\ \mathcal{C}_R \in \mathcal{S}\big] = \mathbb{E}\big[\Phi_a(C_R);\ R \nleftrightarrow a,\ \mathcal{C}_R \in \mathcal{S}\big]\,.$$

If moreover there is a vertex s in R with $\mathbb{E}\big[F(C_a)\big] \leq \mathbb{E}\big[F(C_s)\big]$, then $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a] \geq 0$.

Proof. For the identity, sum over the values of $\mathcal{C}_R$. On $\{R \nleftrightarrow a\}$ the vertex a lies outside C_R, every pair with exactly one endpoint in C_R is closed and the pairs with both endpoints outside C_R retain their original weights, so the conditional mean of $F(C_a)$ is $F(C_R) - \Phi_a(C_R)$.

For the inequality, assume $a \neq s$, since otherwise $\{R \nleftrightarrow a\}$ is empty by reflexivity. Let $\pi(K)$ be the probability that no vertex of $R\backslash\{s\}$ reaches a in a configuration in which every pair meeting $\{s\}\cup\operatorname{span} K$ is closed and every remaining pair keeps its original weight. Enlarging K deletes more pairs, so π is nonnegative and increasing. Summing over the values of $\mathcal{C}_s$ gives, for every real function p of sets of pairs,

$$\mathbb{E}\big[p(\mathcal{C}_s);\ R \nleftrightarrow a\big] = \mathbb{E}\big[p(\mathcal{C}_s)\pi(\mathcal{C}_s);\ s \nleftrightarrow a\big]\,,$$

because on $\{s \nleftrightarrow a\}$ a vertex of R inside C_s misses a automatically, while for a vertex outside C_s both its cluster and C_a are computed in that configuration. Apply (6) with the vertex s, with $X = \{a\}$ and with the increasing functions $p(\mathcal{C}) = \Phi_a(\{s\} \cup \operatorname{span}\mathcal{C})$ and π:

$$\mathbb{E}\big[\Phi_a(C_s);\ s \nleftrightarrow a\big]\,\mathbb{E}\big[\pi(\mathcal{C}_s);\ s \nleftrightarrow a\big] \le \mathbb{P}(s \nleftrightarrow a)\,\mathbb{E}\big[\Phi_a(C_s);\ R \nleftrightarrow a\big]\,.$$

The first factor on the left is $\mathbb{E}\big[F(C_s)\big] - \mathbb{E}\big[F(C_a)\big] \ge 0$ by Lemma 5.1(ii) and the second is nonnegative, so if $\mathbb{P}(s \nleftrightarrow a) > 0$ then $\mathbb{E}[\Phi_a(C_s); R \nleftrightarrow a] \ge 0$, while if $\mathbb{P}(s \nleftrightarrow a) = 0$, then the event $\{R \nleftrightarrow a\}$ is null. Since Φ_a is increasing and $C_s \subseteq C_R$, the identity of the first part with $\mathcal{S}$ containing every set of pairs gives $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a] = \mathbb{E}[\Phi_a(C_R); R \nleftrightarrow a] \ge \mathbb{E}[\Phi_a(C_s); R \nleftrightarrow a] \ge 0$. □

Proof of Theorem 7.1. Put $T = A \setminus \{a\}$. If $a \in R$ then $\{R \nleftrightarrow a\}$ is empty by reflexivity and both sides of (22) vanish. Suppose next that R meets T, and choose a vertex s in $R \cap T$. Then $\{R \leftrightarrow T\}$ is the sure event, so the left side of (22) is $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a]$, and since $s \in A$ the hypothesis gives $\mathbb{E}\big[F(C_a)\big] \le \mathbb{E}\big[F(C_s)\big]$, so Lemma 7.17 applies. In the remaining case R is disjoint from A.

Let T_1 be the set of $x \in T$ with $\mathbb{P}(x \nleftrightarrow a) > 0$. Apply Theorem 7.16 with $Y = \{a\}$, with the set T_1, with the increasing function Φ_a and with the set R, which is disjoint from $\{a\} \cup T_1$. By Lemma 5.1(ii) the number (13) attached to $x \in T_1$ is $(\mathbb{E}\big[F(C_x)\big] - \mathbb{E}\big[F(C_a)\big])/\mathbb{P}(x \nleftrightarrow a) \ge 0$, so the subtracted sum is nonnegative and

$$\mathbb{E}\big[\Phi_a(C_R);\ R \nleftrightarrow a,\ R \leftrightarrow T_1\big] \ge 0\,.$$

The event $\{R \leftrightarrow T_1\}$ is determined by $\mathcal{C}_R$, so the first part of Lemma 7.17 converts this into $\mathbb{E}[F(C_R) - F(C_a); R \nleftrightarrow a, R \leftrightarrow T_1] \ge 0$. Finally, a configuration in $\{R \nleftrightarrow a\} \cap \{R \leftrightarrow T\}$ but not in $\{R \leftrightarrow T_1\}$ has R joined to at least one vertex x of $T \setminus T_1$ while missing a, hence lies in the null event $\{x \nleftrightarrow a\}$; a finite union of null events is null, so the two set integrals agree and (22) follows. □

8 Deleting the Edges at the Starting Vertex

Question 2.9 selects the vertex a by a probability computed with every pair incident to o closed, and then asks for (4) in the original measure. Exposing the pairs at o turns the cluster of o into the union of the clusters of a random set of vertices, and Theorem 7.1 applies to each value of that set.

Theorem 8.1. *Let A be a nonempty finite set of vertices, let o and b be vertices, and let $a \in A$ minimise $\mathbb{P}_\star(x \leftrightarrow b)$ over $x \in A$, where $\mathbb{P}_\star$ is the percolation measure in which every pair incident to o is closed and all other weights are unchanged. Then* (4) *holds. In particular Question 2.9 has an affirmative answer.*

Proof. Condition on the states of the pairs incident to o. For such a state ζ let R_ζ be the union of $\{o\}$ and the set of vertices $u \neq o$ with $\{o, u\}$ open in ζ. The pairs not incident to o have exactly their law under $\mathbb{P}_\star$, and they are independent of ζ. Under $\mathbb{P}$ a path leaving o starts with a pair of ζ and continues among the pairs not incident to o, possibly returning to o and leaving again, so, pointwise in the remaining configuration,

$$C_o = C_{R_\zeta}, \qquad \{o \leftrightarrow A\} = \{R_\zeta \leftrightarrow A\}\,, \tag{40}$$

the right sides being computed under $\mathbb{P}_\star$, for which o is isolated and therefore contributes the single vertex o to the union. Moreover, on $\{o \nleftrightarrow a\}$ no path from a can reach o, so no such path uses a pair of ζ, and C_a is the same under both measures.

Put $F = \mathbf{1}\{b \in \cdot\}$. On $\{o \leftrightarrow a\}$ the clusters of o and a coincide, so

$$\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) - \mathbb{P}(a \leftrightarrow b, o \leftrightarrow A) = \mathbb{E}\big[F(C_o) - F(C_a);\ o \nleftrightarrow a,\ o \leftrightarrow A \setminus \{a\}\big]\,,$$

and by (40) the right side equals

$$\sum_\zeta \mathbb{P}(\zeta)\,\mathbb{E}_\star\big[F(C_{R_\zeta}) - F(C_a);\ R_\zeta \nleftrightarrow a,\ R_\zeta \leftrightarrow A \setminus \{a\}\big]\,,$$

the events and clusters inside the expectation being computed under $\mathbb{P}_\star$. The hypothesis on a says $\mathbb{E}_\star\big[F(C_a)\big] \le \mathbb{E}_\star\big[F(C_x)\big]$ for every $x \in A$, so each summand is nonnegative by Theorem 7.1 applied under $\mathbb{P}_\star$ with the set R_ζ. Averaging gives (4). No division is used, so states ζ of probability zero and coincidences among o, a, b and the vertices of A are covered. □

9 A Four-Vertex Example, and Sets of Two Vertices

Question 2.8 selects the vertex a by the probability of the joint event that a reaches b and that o misses A. Write ρ_x for that probability, so that

$$\rho_x = \mathbb{P}(x \leftrightarrow b,\ o \nleftrightarrow A) \qquad \text{for } x \in A\,.$$

When several vertices of A realise the minimum of ρ, the reading in which every one of them satisfies (4) is false.

Proposition 9.1. *There are a graph with four vertices, weights in $[0,1]$, a set A with two elements and vertices o and b outside A such that both elements of A minimise ρ while (4) fails for one of them.*

Proof. Let $V = \{o, a_1, a_2, b\}$ and $A = \{a_1, a_2\}$, give the pairs $\{o, a_1\}$ and $\{a_2, b\}$ weight one and the pair $\{a_1, a_2\}$ weight one half, and give every other pair weight zero. Since $\{o, a_1\}$ is open with probability one and $a_1 \in A$, the event $\{o \leftrightarrow A\}$ has probability one, so $\rho_{a_1} = \rho_{a_2} = 0$ and both vertices of A minimise. The event $\{a_2 \leftrightarrow b\}$ has probability one, so

$$\mathbb{P}(o \leftrightarrow A,\ a_2 \leftrightarrow b) = 1\,.$$

The event $\{o \leftrightarrow b\}$ requires the pair $\{a_1, a_2\}$ to be open, so $\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A) = \frac{1}{2}$. Thus (4) fails for $a = a_2$. □

The minimised quantity vanishes at both vertices of A in this example, and nothing like it can happen when A has at most two elements: Theorem 9.4 settles both readings there. Two preparations are needed. The first removes from (4) the part of the event $\{o \leftrightarrow A\}$ on which its two sides agree.

Lemma 9.2. *Let A be a nonempty finite set of vertices, let $a \in A$, and let b and o be vertices. Then*

$$\mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b) - \mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) = \mathbb{P}\big(a \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow A \setminus \{a\}\big) - \mathbb{P}\big(o \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow A \setminus \{a\}\big)\,.$$

Proof. Split $\{o \leftrightarrow A\}$ into $\{o \leftrightarrow a\}$ and $\{o \nleftrightarrow a\} \cap \{o \leftrightarrow A\}$, the second of which is $\{o \nleftrightarrow a\} \cap \{o \leftrightarrow A \setminus \{a\}\}$. On $\{o \leftrightarrow a\}$ the open clusters of o and a coincide, so the events $\{a \leftrightarrow b\}$ and $\{o \leftrightarrow b\}$ agree there and the two contributions from that part cancel. □

The second preparation is a four-product form of Theorem 3.2, in which two functions of one kind are exchanged against two functions of the other.

Lemma 9.3. *Let S_1 and S_2 be sets of vertices, let g_3 and g_4 be nonnegative real functions of two sets of pairs, each increasing in the first argument and decreasing in the second, and let h_1 and h_2 be nonnegative real functions of two sets of pairs, each decreasing in the first argument and increasing in the second. Then, with every function evaluated at $(\mathcal{C}_{S_1}, \mathcal{C}_{S_2})$ and every expectation restricted to the event $\{S_1 \nleftrightarrow S_2\}$,*

$$\mathbb{E}[g_3h_1]\,\mathbb{E}[g_4h_2] \le \mathbb{E}[g_3g_4]\,\mathbb{E}[h_1h_2]\,.$$

Proof. Write D for the event $\{S_1 \nleftrightarrow S_2\}$. If $\mathbb{P}(D) = 0$, then both sides vanish, so assume $\mathbb{P}(D) > 0$. Theorem 3.2 applies to the pair g_3, g_4 and, after exchanging S_1 and S_2, to the pair h_1, h_2, giving

$$\mathbb{E}[g_3]\,\mathbb{E}[g_4] \le \mathbb{P}(D)\,\mathbb{E}[g_3g_4], \qquad \mathbb{E}[h_1]\,\mathbb{E}[h_2] \le \mathbb{P}(D)\,\mathbb{E}[h_1h_2]\,,$$

while (5) applies to the pair g_3, h_1 and to the pair g_4, h_2, giving

$$\mathbb{P}(D)\,\mathbb{E}[g_3h_1] \le \mathbb{E}[g_3]\,\mathbb{E}[h_1], \qquad \mathbb{P}(D)\,\mathbb{E}[g_4h_2] \le \mathbb{E}[g_4]\,\mathbb{E}[h_2]\,.$$

All eight quantities are nonnegative, so the two inequalities of the second display may be multiplied and the result bounded by the product of the two inequalities of the first:

$$\mathbb{P}(D)^2\,\mathbb{E}[g_3h_1]\,\mathbb{E}[g_4h_2] \le \mathbb{E}[g_3]\,\mathbb{E}[h_1]\,\mathbb{E}[g_4]\,\mathbb{E}[h_2] \le \mathbb{P}(D)^2\,\mathbb{E}[g_3g_4]\,\mathbb{E}[h_1h_2]\,.$$

Divide by $\mathbb{P}(D)^2$. $\square$

Theorem 9.4. *Let o and b be vertices and let A be a nonempty set of vertices with at most two elements. If $a \in A$ is the unique vertex of A realising the minimum of ρ, then (4) holds. If moreover $\mathbb{P}(o \nleftrightarrow A) > 0$, then (4) holds for every $a \in A$ realising that minimum.*

Proof. We argue in four steps: a set with one element; a set with two elements at which the minimum is strict; the first assertion; and the tied minima of the second assertion.

We start with a set with one element. If $A = \{a\}$ then on $\{o \leftrightarrow a\}$ the open clusters of o and a coincide, so $\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) = \mathbb{P}(o \leftrightarrow a, o \leftrightarrow b) = \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$, and both assertions hold with equality.

Next we treat $A = \{a, x\}$ with $\rho_a < \rho_x$. By Lemma 9.2, and because the open clusters of o and x coincide on $\{o \leftrightarrow x\}$, the inequality (4) to be proved reads

$$\mathbb{P}(a \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow x) \le \mathbb{P}(x \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow x)\,. \tag{41}$$

Apply Lemma 9.3 with $S_1 = \{a\}$ and $S_2 = \{o, x\}$, and write D for the event $\{a \nleftrightarrow o,\ a \nleftrightarrow x\}$. Take for g_3 and g_4 the indicators of $\{a \leftrightarrow b\}$ and of $\{o \nleftrightarrow x\}$, and for h_1 and h_2 the indicators of $\{o \leftrightarrow x\}$ and of $\{x \leftrightarrow b\}$. The event $\{a \leftrightarrow b\}$ is determined by the open pairs lying on paths from a and increases with them, while the events $\{o \leftrightarrow x\}$ and $\{x \leftrightarrow b\}$ are determined by the open pairs lying on paths from $\{o, x\}$ and increase with them; adding pairs to the second of these sets can only merge components. Hence g_3 and g_4 are increasing in the first argument and decreasing in the second, and h_1 and h_2 are decreasing in the first and increasing in the second.

We identify the four expectations. On $\{o \leftrightarrow x\}$ the open clusters of o and x coincide, so $a \nleftrightarrow o$ and $a \nleftrightarrow x$ are the same condition there and $D \cap \{o \leftrightarrow x\} = \{o \nleftrightarrow a,\ o \leftrightarrow x\}$; therefore

$$\mathbb{E}[g_3h_1; D] = \mathbb{P}(a \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow x), \qquad \mathbb{E}[h_1h_2; D] = \mathbb{P}(x \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow x)\,.$$

On the other hand $D \cap \{o \nleftrightarrow x\} = \{o \nleftrightarrow A\} \cap \{a \nleftrightarrow x\}$, and two vertices joined to b are joined to each other, so on that event $\{a \leftrightarrow b\}$ forces $x \nleftrightarrow b$ and $\{x \leftrightarrow b\}$ forces $a \nleftrightarrow b$, and conversely. Therefore

$$\mathbb{E}[g_3g_4; D] = \mathbb{P}(o \nleftrightarrow A,\ a \leftrightarrow b,\ x \nleftrightarrow b), \qquad \mathbb{E}[g_4h_2; D] = \mathbb{P}(o \nleftrightarrow A,\ x \leftrightarrow b,\ a \nleftrightarrow b)\,,$$

and Lemma 9.3 reads

$$\begin{aligned}\mathbb{P}(a \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow x)\,&\mathbb{P}(o \nleftrightarrow A,\ x \leftrightarrow b,\ a \nleftrightarrow b)\\ &\le\ \mathbb{P}(o \nleftrightarrow A,\ a \leftrightarrow b,\ x \nleftrightarrow b)\,\mathbb{P}(x \leftrightarrow b,\ o \nleftrightarrow a,\ o \leftrightarrow x).\end{aligned} \tag{42}$$

Splitting $\{o \nleftrightarrow A,\ a \leftrightarrow b\}$ and $\{o \nleftrightarrow A,\ x \leftrightarrow b\}$ according as the other vertex of A reaches b,

$$\rho_a - \mathbb{P}(o \nleftrightarrow A,\ a \leftrightarrow b,\ x \nleftrightarrow b) = \mathbb{P}(o \nleftrightarrow A,\ a \leftrightarrow b,\ x \leftrightarrow b) = \rho_x - \mathbb{P}(o \nleftrightarrow A,\ x \leftrightarrow b,\ a \nleftrightarrow b)\,,$$

so $\rho_a < \rho_x$ gives $\mathbb{P}(o \nleftrightarrow A, a \leftrightarrow b, x \nleftrightarrow b) < \mathbb{P}(o \nleftrightarrow A, x \leftrightarrow b, a \nleftrightarrow b)$, and in particular the second of these two numbers is positive. Bounding the first factor on the right of (42) by the second number and cancelling that positive factor gives (41).

For the first assertion, a set with one element is covered by the first step. If $A = \{a, x\}$ and a is the unique vertex realising the minimum, then $\rho_a < \rho_x$ and the second step applies.

For the second assertion, assume $\mathbb{P}(o \nleftrightarrow A) > 0$ and let a realise the minimum. Again only $A = \{a, x\}$ has to be treated, and only $\rho_a = \rho_x$, the case $\rho_a < \rho_x$ being the second step.

Suppose first that $\rho_a = 0$. A configuration has positive probability precisely when it contains every pair of weight one and no pair of weight zero. The event $\{o \nleftrightarrow A\}$ is decreasing and has positive probability, so it contains the configuration whose open pairs are exactly those of weight one; that is, the component of o in the graph of the pairs of weight one misses A. Suppose that $\mathbb{P}(a \leftrightarrow b,\ o \nleftrightarrow b) > 0$ and choose a configuration of positive probability in which $a \leftrightarrow b$ and $o \nleftrightarrow b$, together with a simple open path from a to b in it. Let ω be the configuration whose open pairs are those of weight one together with the pairs of that path. Then ω has positive probability, it is contained in the chosen configuration, and $a \leftrightarrow b$ in ω. Moreover $o \nleftrightarrow A$ in ω: a path in ω from o to a vertex of A either uses only pairs of weight one, which contradicts the preceding sentence about the component of o, or meets the chosen path, in which case $o \leftrightarrow b$ in ω and hence in the chosen configuration, contrary to its choice. Thus $\rho_a > 0$, a contradiction. Hence $\{a \leftrightarrow b\}$ is contained in $\{o \leftrightarrow b\}$ up to a null event, and when both hold the vertex o is joined to a through b, so $\{o \leftrightarrow A\}$ holds as well; therefore $\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) \le \mathbb{P}(a \leftrightarrow b) \le \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$.

Suppose finally that $\rho_a > 0$, and fix q with $0 < q < 1$. Enlarge the vertex set by one vertex y, give the pair $\{y, a\}$ weight q and every other pair at y weight zero, leave every old weight unchanged, and replace A by $\{y, x\}$; write $\mathbb{P}_q$ for the enlarged measure. Condition on the state of $\{y, a\}$. If that pair is open then y is joined exactly to the vertices of the open cluster of a, the connections among the old vertices are unchanged, and the events $\{o \leftrightarrow \{y, x\}\}$ and $\{y \leftrightarrow b\}$ are $\{o \leftrightarrow A\}$ and $\{a \leftrightarrow b\}$. If it is closed then y is isolated, so $\{y \leftrightarrow b\}$ is empty and $\{o \leftrightarrow \{y, x\}\}$ is $\{o \leftrightarrow x\}$. Hence

$$\mathbb{P}_q(y \leftrightarrow b,\ o \nleftrightarrow \{y, x\}) = q\rho_a, \qquad \mathbb{P}_q(x \leftrightarrow b,\ o \nleftrightarrow \{y, x\}) = q\rho_x + (1-q)\mathbb{P}(x \leftrightarrow b,\ o \nleftrightarrow x)\,,$$

$$\mathbb{P}_q(o \leftrightarrow \{y, x\},\ y \leftrightarrow b) = q\,\mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b)\,,$$

$$\mathbb{P}_q(o \leftrightarrow b,\ o \leftrightarrow \{y, x\}) = q\,\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) + (1-q)\,\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow x)\,.$$

Since $\{o \nleftrightarrow A\}$ is contained in $\{o \nleftrightarrow x\}$, the number $\mathbb{P}(x \leftrightarrow b, o \nleftrightarrow x)$ is at least ρ_x, so the second of the two numbers in the first of these displays is at least $\rho_x = \rho_a > q\rho_a$, which is the first: the vertex y is the unique vertex of $\{y, x\}$ realising the minimum, in a set with two elements. By the first assertion,

$$q\,\mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b) \le q\,\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) + (1-q)\,\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow x)\,.$$

Suppose that $\delta = \mathbb{P}(o \leftrightarrow A, a \leftrightarrow b) - \mathbb{P}(o \leftrightarrow b, o \leftrightarrow A)$ is positive; then $\delta \le 1$, and the last display gives $q\delta \le (1-q)\mathbb{P}(o \leftrightarrow b, o \leftrightarrow x) \le 1 - q$ for every q with $0 < q < 1$. Taking $q = 1 - \delta/4$, which lies between $3/4$ and 1, turns this into $(1 - \delta/4)\delta \le \delta/4$, that is $\delta \ge 3$, a contradiction. Hence $\delta \le 0$, which is (4). $\square$

Remark 9.5. Whether (4), the bound comparing $\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A)$ with $\mathbb{P}(o \leftrightarrow A,\ a \leftrightarrow b)$, holds for the unique vertex realising the minimum of ρ when A has three or more elements is open. If instead a is chosen to minimise $\mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$ over $x \in A$, then (4) holds by (1).

10 The Lean Formalization

Anthropic's Lean proof of Conjecture 2.3, written in Lean 4 with Mathlib [Dev26], also kernel-checks part of what is proved above. Its own main theorem is Theorem 3.3, formalized as `Percolation.Continuity.CSH.cshHolds`. On top of that theorem, the Lean proof of Conjecture 2.3 kernel-checks Theorems 4.1 and 5.2, Corollaries 5.4 and 5.5, Question 2.6 with the right side printed by Kozma and Nitzan, Theorem 7.1 for a set $R = \{v, w\}$ of at most two vertices, the comparison (23) and Conjecture 2.5, the reduction of Theorem 8.1 to Theorem 7.1 for an arbitrary finite set, and Proposition 9.1. The declarations are `KNAll.genY_all`, `KNAll.conjecture4Fixed_holds`, the declarations of `Conjectures.lean` and `ClusterProperty.lean`, `KNAll.question5_holds`, `KNAll.Guarded.pair_fixedMin`, `KNAll.Guarded.conjecture6Strong_holds`, `KNAll.Guarded.conjecture6_holds`, `KNAll.question9_of_setSourceFixedMin` and `KNAll.not_question8EveryMin`. Each of them compiles under Lean `4.32.0` with Mathlib commit `81a5d257c8e410db227a6665ed08f64fea08e997`, with no `sorry`, and depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Four results above go beyond that. The strengthened right side of Theorem 6.2, Theorems 7.1 and 7.16 for an arbitrary finite set of vertices, and Theorem 8.1 without a hypothesis are proved in this document and are not closed Lean theorems. In particular `KNAll.question9_of_setSourceFixedMin` assumes Theorem 7.1 for an arbitrary finite set and does not prove it.

All of this was done autonomously by ChatGPT 5.6 Sol and Claude Fable 5.1, with minimal human intervention.

References

[AD78] R. Ahlswede and D. E. Daykin, *An inequality for the weights of two families of sets, their unions and intersections*, Z. Wahrscheinlichkeitstheorie verw. Gebiete **43** (1978), 183–185.

[BHK06] J. van den Berg, O. Häggström and J. Kahn, *Some conditional correlation inequalities for percolation and related processes*, Random Structures Algorithms **29** (2006), 417–435. Theorem 1.1 (p. 3) and Theorem 2.1 (p. 9) with $q = 1$, the latter being Theorem 1.5 (p. 7) read with Remark 1 (p. 5), are Theorem 3.1 and Theorem 3.2.

[Dev26] *Theta at the critical point vanishes for Bernoulli bond percolation on the hypercubic lattice in all dimensions at least two, in Lean 4 and Mathlib*, 2026, `https://github.com/anthropic-experimental/percolation`.

[H60] T. E. Harris, *A lower bound for the critical probability in a certain percolation process*, Proc. Cambridge Philos. Soc. **56** (1960), 13–20.

[KN24] G. Kozma and S. Nitzan, *A reduction of the percolation at criticality problem to a conjectured inequality*, arXiv:2401.12397, 2024. Conjectures 1 to 3 with display (3) are on pp. 3 and 15, Conjecture 4 and Question 5 on pp. 31–32, Conjecture 6 on p. 34, and Questions 7 to 9 with display (41) on p. 36.